40th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit
11 - 14 July 2004, Fort Lauderdale, Florida

AIAA Paper 2004-4168

Combustor Diffuser Modeling
Part I: Inlet Profiles & 2-D Calculations

H. Mongia[C], G. Hsiao, and D. Burrus, GE Aircraft Engines, Cincinnati,
PSVS Sreedhar, A. Rao, and P. Naik, EACoE, Bangalore

ABSTRACT

2-D axisymmetric simulation of a typical gas turbine diffusion system is conducted with two turbulence models, the standard k-ε, Shear Stress Turbulence, SST k-ω including its low-Reynolds number formulation. Effect of diffuser inlet profile, center-, inner- and outer-peaked on pressure recovery and loss coefficient is modeled with two approaches for specifying the inlet conditions: 1) Measured inlet mean axial velocity with reasonable assumptions for turbulence quantities; 2) Through-the-rig inlet, profiler, duct, prediffuser and dump region 3-D CFD simulation to calculate the pre-diffuser inlet profiles. Best agreement with data is obtained with the standard k-ε model starting with the calculated diffuser inlet conditions.

INTRODUCTION

2-D and 3-D CFD calculations are generally done during the gas turbine combustion system design process to aid in the optimization of the diffusion system. Early work by Karki et al[1] as well as the follow-on publications, e.g., Reference 2 have shown that the standard k-ε model with wall functions overpredicts C_P by 5-15% compared to data. A preliminary assessment of various advanced turbulence models indicates[3] that SST k- ω turbulence model has shown potential for giving better agreement with data for flows subject to adverse pressure gradients including separation. However due to lack of systematic validation of CFD-based approach with data of combustor diffusion systems of practical interest to designers, the current design process relies on empirical data collected on simulated full-scale 3-D diffuser rigs coupled with qualitative guidance from CFD.

Let us review a typical 3-D diffuser experimental investigation in order to establish an outline of what CFD-based diffuser model validation effort needs to undertake so that "CFD-friendly combustor designers" don't need to rely on diffuser rig testing as part of the design process.

Karki et al[1] made a significant contribution to the art of combustor diffuser modeling even though there still remained several unresolved issues due mainly to:

1. Mean square root error, σ of measured C_P was of the order of 10% since the data was from a full-annular combustor rig, which is normally not as well instrumented and controlled as typical special purpose diffuser rig is.
2. No total pressure loss coefficient, Ω data was available for model validation.
3. Limited 2-D grid refinement investigations were undertaken. 3-D calculations with coarse-grid arrangement were made to illustrate that practical combustor diffusers need to employ 3-D CFD techniques before reaching any definite conclusions about the standard k-ε model's capabilities in this area.
4. Data was available only on one baseline combustion system with only one prediffuser inlet profile and a set level of airflow splits around the combustor.

Prompted by recent advances in CFD simulation technology, turbulence modeling and availability of an excellent data set, the designers' need to undertake integrated combustor liner and diffuser modeling[4], we have undertaken a comprehensive investigation to study the effect of the following on C_P and Ω:

1. Diffusion system design parameters including inlet profiles, pre-diffuser area ratio, L/H, cowling geometry, blockages due to dome swirlers, nozzle stem and tip, airflow splits between the dome, inner and outer annuli of the combustor liner.
2. Procedures for calculating diffuser inlet conditions.
3. 2-D versus 3-D CFD simulations.
4. Different turbulence models and near-wall treatments to capture boundary layer losses and flow separation.

This paper, Part I, focuses on the procedures for calculating the inlet profiles specific to Runs 3, 4 and 5 of Table 1 and determine how far one can go with 2-D CFD simulations, the standard k-ε model along with the

C: Copyright © 2004 by the authors. Published by AIAA with permission.

Copyright © 2004 by The authors. Published by the American Institute of Aeronautics and Astronautics, Inc., with permission.

two near-wall treatments, and SST k-ω turbulence model coded in FLUENT® software.

DATA SETS

A typical annular diffuser rig, Figure 1a, consists of a cylindrical annular inlet section, which extended from a rounded entrance to the diffuser inlet. The length of this inlet section from the bell mouth entrance to the prediffuser inlet was ~12.0"or ~16 passage heights. The axial distance from the inlet section entrance to the diffuser inlet measurement plane was ~ 8.7" or ~11 passage heights. The distance from the measurement plane to the pre-diffuser inlet was ~3.2" or ~4 passage heights. Within the inlet section at a distance of ~2.5" downstream from the rounded entrance were located the inlet profilers. In the initial rig configuration the profiles consisted of a pair of 360-degree rings designed to protrude 0.10" into the flow stream from both the inner and outer flow path walls, called opposed-ring profiler. The location of these rings placed them ~12 passage heights upstream of the diffuser inlet. The testing reported in this paper used "bow tie" profilers, Fig. 1b. The three shown produce center-, inner- and outer-peak profiles, respectively.

The prediffuser exit angle, exit height, L/H and geometric area are ~10.25^{O}, 1.17", 2.27 and 74.8 in^2, respectively. Cowl opening height, H_{cowl}, geometric area, axial location, L_{cowl} and $(L/H)_{cowl}$ are 0.729", 100 in^2, 2.125" and 2.92, respectively. Outer and inner annuli areas at stations 3.5 (toward the farther end of the annulus) are 39.4 and 30.5 in^2, respectively. The airflow splits between the three streams (dome, inner and outer annuli) are controlled independently in order to evaluate the effect of the flow splits on C_P and Ω.

A total of 118 static and total pressure taps were installed on the diffuser model test rig, shown schematically in Fig. 1a. Of these 64 were located in the rig inlet and prediffuser sections, the remaining 54 were located in the dump regions, the combustor dome, and upstream of the metering orifices. As part of the inlet instrumentation there were four 5-element total pressure rakes along with wall static pressures for each rake. The elements of these rakes were located at y/h = 0.1033, 0.3081, 0.5095, 0.7078, and 0.9031, bottom to top. At the diffuser discharge plane another set of four 5-element total pressure rakes along with associated wall statics were located. The radial positions of these total pressure elements were at y/h = 0.1045, 0.3136, 0.5227, 0.7365, and 0.9233, bottom to top. As part of the combustor dome instrumentation two mixer cups were each instrumented with five total pressures and five static pressures equally spaced around the cup circumferences. These taps were placed to quantify the pressure variation around the mixer cups. Pt3 is based on the calculated value determined from the measured diffuser inlet static pressure and the known inlet mass flow rate, assuming C_D of 1.0. The dome total pressure is taken from the average of the dome pressures mounted onto the dome plate facing upstream, assuming stagnation flow conditions.

For the selected series of 3-D diffuser rig testing to be compared with predictions in this paper and References 7-9, a total of 10 good quality runs (Table 1) are used to quantify the effect of the following on pressure recovery and total pressure loss coefficients, C_P and Ω:

1. Diffuser inlet profiles, Runs 3, 4, 5 and 8 Tables 2 and 3;
2. Diffuser area ratio and length, Runs 3 and 9;
3. W/ and w/o combustor dome cowling, Runs 3 and 6;
4. Cowling inlet area, Runs 3 and 7;
5. Nozzle tip diameter, Runs 3 and 11;
6. Blockage due to nozzle stem, Runs 3 and 12;
7. Airflow splits between dome, inner and outer passages that was conducted for all the 10 test cases.

Here static pressure recovery and total pressure loss coefficients C_P and Ω are:

$$C_P = (P_s - P_{s3})/q_3$$

$$\Omega = (P_{t3} - P_t)/q_3$$

Subscripts s, t and 3 denote static and total pressures, combustor inlet station, respectively; q_3 is the diffuser inlet dynamic head.

Tables 2 and 3 list the average C_P, total pressure loss coefficients, Ω and their root mean square error (σ) values for each of the 10 runs at the pre-diffuser exit plane. Each run included a set of 18 test conditions involving variations in airflow splits between the three streams, namely outer passage, inner passage and the dome. As noted in Table 1, Run #10 was a repeat of Run #3 for establishing typical test-to-test variability. From this we observe the following:

1. Sum of root mean squares, so-called pool σ for C_P and Ω at the pre-diffuser exit of 2.08% and 8.05% are comparable with the test-to-test variability of 2.64% and 9.71%.
2. On the other hand, the max to min variation in C_P for the test Runs 3, 4 and 5 (conducted to quantify the effect of inlet profile) is 13.3%, which is ~5 times the test to test variability σ or ~6.4 times the pool σ. The overall σ for the 10 runs is ~8% or 4 times the pool sigma. Essentially similar ratios may be observed for Ω.
3. It is a reasonable assumption to make that the pool σ's for the dome, inner and outer passages should be ≥ those of the diffuser exit plane.

4. Therefore, consistent with our long-term objectives[4] for pretest "no-fudge factor" prediction capability for combustor design, the CFD-based diffusion system design tools should match C_P and Ω data for each design options (e.g., 9 in this experimental investigation) with σ of 2.1% and 8.1%, respectively. Moreover, the models should have enough credibility with the designers that the predicted σ variations of C_P and Ω between the nine configurations of the orders of 9.1% and 27.4%, respectively can be believed, and consequently there would be no need to run a special purpose diffuser rig test as a part of the design process.

We would like to achieve agreement for CP and Ω data within root-mean square errors of ~5% and 15%, respectively.

CALCULATIONS PROCESS

A commercially available code, FLUENTR CFD was used in our investigation since it already has the turbulence models we had investigated in our earlier studies, e.g., References 1 and 3, in addition to having an excellent griding strategy suitable for the test configuration of interest to us. The reader should refer to FLUENT® software documentation for details on the two turbulence models and the two types of wall functions, (standard k-ε and shear stress transport SST k-ω models; standard and non-equilibrium wall functions) used in this study. Brief description is given in the following paragraphs.

The standard k-ε model is a semi-empirical model based on model transport equations for the turbulence kinetic energy (k) and its dissipation rate (ε). The model transport equation for k is derived from the exact equation while the model equation for ε is obtained using physical reasoning. The transport equations for k and ε used in the standard k-ε model are:

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}[(\mu+\frac{\mu_t}{\sigma_k})\frac{\partial k}{\partial x_j}]+G_k+G_b-\rho\varepsilon-Y_m$$

$$\frac{\partial}{\partial t}(\rho\varepsilon)+\frac{\partial}{\partial x_i}(\rho\varepsilon u_i)=\frac{\partial}{\partial x_j}[(\mu+\frac{\mu_t}{\sigma_k})\frac{\partial\varepsilon}{\partial x_j}]+C_{1\varepsilon}\frac{\varepsilon}{k}(G_k+C_{3\varepsilon}G_b)-C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}$$

G_k represents the generation of turbulence kinetic energy due to mean velocity gradients and is given by,

$$G_k=-\rho u_i' u_j'\frac{\partial u_j}{\partial x_i}$$

G_b represents the generation of turbulence kinetic energy due to buoyancy and is given by,

$$G_k=\beta g_i\frac{\mu_t}{\Pr_t}\frac{\partial T}{\partial x_i}$$

Where Pr_t is the turbulent Prandtl number for energy, g_i is the component of gravitational vector in the ith direction and β is the coefficient of thermal expansion.
Y_m represents the contribution fluctuating dilatation in compressible turbulence to the overall dissipation rate and is given by,

$$Y_m=2\rho\varepsilon M_t^2$$

Where M_t is the turbulent Mach number, defined as

$M_t=\sqrt{\frac{k}{\alpha^2}}$ Where α is the speed of sound.

$C_{1\varepsilon}$, $C_{2\varepsilon}$, $C_{3\varepsilon}$ are constants and σ_k, σ_ε are the turbulent Prandtl numbers for k and ε. The turbulent viscosity μ_t is calculated using k and ε:

$\mu_t=\rho C_\mu\frac{k^2}{\varepsilon}$ Where C_μ is a constant.

In this study the following values for these constants were used:
$C_{1\varepsilon}=1.44$, $C_{2\varepsilon}=1.92$, $C_\mu=0.09$, $\sigma_k=1.00$, $\sigma_\varepsilon=1.30$.

The standard wall functions in FLUENT® software are based on proposition of Launder and Spalding[6] and have been most widely used for industrial applications.

The law of the wall for mean velocity yields:

$$U^*=\frac{1}{\kappa}\ln(Ey^*)$$

where, $U^*=\frac{U_p C_\mu^{1/4}k_p^{1/2}}{\tau_w/\rho}$

$$y^*=\frac{\rho C_\mu^{1/4}k_p^{1/2}y_p}{\mu}$$

and κ = von Karman constant (0.42)
E = empirical constant (9.81)
U_p = mean velocity at point P near the wall.
k_p = turbulence kinetic energy at point P.
y_p = distance from point P to the wall
μ = dynamic viscosity of the fluid.

The logarithmic law for mean velocity is used when $y^* > 11.225$. When the grid is such that $y^* < 11.225$ the laminar stress-strain relationship $U^* = y^*$ is used.

The production of kinetic energy, G_k, and its dissipation rate ε, at the wall adjacent cells, which are the

source terms in the k equation are computed on the basis of the local equilibrium hypothesis. Under this hypothesis, the production of k and its dissipation rate are assumed to be equal in the wall adjacent control volume.

Thus the production of k is given by,

$$G_k \approx \tau_w \frac{\partial U}{\partial y} = \tau_w \frac{\tau_w}{\kappa \rho C_\mu^{1/4} k_p^{1/2} y_p}$$

and ε is computed from

$$\varepsilon_P = \frac{C_\mu^{3/4} k_p^{3/2}}{\kappa y_p}$$

There are two key differences in the non-equilibrium wall functions as compared to the standard wall function:

1. Launder and Spalding's log law for mean velocity is sensitized to pressure-gradient effects.
2. The two layer based concept is adopted to compute the budget of turbulence kinetic energy (G_k, ε) in the wall neighboring cells.

The log-law for mean velocity sensitized to pressure gradients is

$$\frac{\tilde{U} C_\mu^{1/4} k^{1/2}}{\tau_w / \rho} = \frac{1}{\kappa} \ln(E \frac{\rho C_\mu^{1/4} k^{1/2} y}{\mu})$$

where

$$\tilde{U} = U - \frac{1}{2}\frac{dp}{dx}[\frac{y_v}{\rho\kappa\sqrt{k}}\ln(\frac{y}{y_v}) + \frac{y - y_v}{\rho\kappa\sqrt{k}} + \frac{y_v^2}{\mu}]$$

and y_v is the physical viscous sub-layer thickness, and is computed from

$$y_v = \frac{\mu y_v^*}{\rho C_\mu^{1/4} k_p^{1/2}}$$

where $y_v^* = 11.225$.

The non-equilibrium wall function employs the two-layer concept in computing the budget of turbulence kinetic energy at the wall-adjacent cells, which is needed to solve the *k* equation at the wall-neighboring cells. The wall-neighboring cells are assumed to consist of a viscous sub-layer and a fully turbulent layer. The following profile assumptions for turbulence quantities are made:

$$\tau_t = \begin{cases} 0, y < y_v \\ \tau_w, y > y_v \end{cases} \qquad k = \begin{cases} (\frac{y}{y_v})^2 k_p, y < y_v \\ k_p, y > y_v \end{cases}$$

$$\varepsilon = \begin{cases} \frac{2\nu k}{y^2}, y < y_v \\ \frac{k^{3/2}}{C_l y}, y > y_v \end{cases}$$

where $C_l = \kappa C_\mu^{-3/4}$ and y_v is the dimensional thickness of the viscous sub-layer.

Using these profiles the cell averaged production of k for quadrilateral cells for which the volume average can be approximated with a depth-average, is given by

$$\overline{G_k} = \int_0^{y_n} \tau_t \frac{\partial U}{\partial y} dy$$

$$= \frac{1}{\kappa y_n} \frac{\tau_w^2}{\rho C_\mu^{1/4} k_p^{1/2}} \ln(\frac{y_n}{y_v})$$

$$\bar{\varepsilon} = \frac{1}{y_n}\int_0^{y_n} \varepsilon dy = \frac{1}{y_n}[\frac{2\nu}{y_v} + \frac{k_p^{1/2}}{C_l}\ln(\frac{y}{y_n})]k_p$$

Where y_n is the height of the cell ($y_n = 2\, y_p$).

The Shear Stress Transport (SST) *k*-ω model is an empirical model based on model transport equations for the turbulence kinetic energy (*k*) and the specific dissipation rate (ω), which can also be thought of as the ratio of ε to *k*. It is so named because the definition of the turbulent viscosity is modified to account for the transport of the principal turbulent shear stress. Other modifications include the addition of a cross-diffusion term in the ω equation and a blending function to ensure that the model equations behave appropriately in both the near-wall and far-field zones. The transport equations for k and ω used in the SST k-ω model are given below:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j}[\Gamma_k \frac{\partial k}{\partial x_j}] + G_k - Y_k$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega u_i) = \frac{\partial}{\partial x_j}[\Gamma_\omega \frac{\partial \omega}{\partial x_j}] + G_\omega - Y_\omega + D_\omega$$

In these equations, G_k represents the generation of turbulence kinetic energy due to mean velocity gradients and is given by,

$$G_k = -\rho u_i' u_j' \frac{\partial u_j}{\partial x_i}$$

G_ω represents the generation of ω and is given by,

$$G_\omega = -\alpha \frac{\omega}{k} G_k .$$

The coefficient α is given by

$$\alpha = \frac{\alpha_\infty}{\alpha_*}\left(\frac{\alpha_0 + \mathrm{Re}_t / R_\omega}{1 + \mathrm{Re}_t / R_\omega}\right)$$

where, $\alpha_* = \alpha_\infty^* \left(\frac{\alpha_0^* + \mathrm{Re}_t / R_k}{1 + \mathrm{Re}_t / R_k}\right)$.

$$\mathrm{Re}_t = \frac{\rho k}{\mu \omega}, \quad R_k = 6, \quad \alpha_0^* = \frac{\beta_i}{3}, \quad \beta_i = 0.072$$

Γ_k and $\Gamma\omega$ represent the effective diffusivity of k and ω, respectively, which are calculated as described below.

$\Gamma_k = \mu + \frac{\mu_t}{\sigma_k}$ and $\Gamma_\omega = \mu + \frac{\mu_t}{\sigma_\omega}$. where σ_k and σ_ω are the turbulent Prandtl numbers for k and ω, respectively. The turbulent viscosity, μ_t, is computed as follows:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{\max\left[\frac{1}{\alpha^*}, \frac{\Omega F_2}{\alpha_1 \omega}\right]} ;$$

where $\Omega = \sqrt{2\Omega_{ij}\Omega_{ij}}$

$$\sigma_k = \frac{1}{F_1 / \sigma_{k,1} + (1 - F_1) / \sigma_{k,2}]}$$

$$\sigma_\omega = \frac{1}{F_1 / \sigma_{\omega,1} + (1 - F_1) / \sigma_{\omega,2}]}$$

Ω_{ij} is the mean rate-of-rotation tensor.

The blending functions F_1 and F_2, are given by

$$F_1 = \tanh(\Phi_1^4)$$

$$\Phi_1 = \min\left[\max\left(\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right), \frac{4\sigma k}{\sigma_{\omega,2} D_\omega^+ y^2}\right]$$

$$D_\omega^+ = \max\left[\frac{2\rho}{\sigma_{\omega,2}\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}, 10^{-20}\right]$$

$$F_2 = \tanh(\Phi_2^2)$$

$$\Phi_2 = \max\left[\frac{2\sqrt{k}\rho}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right]$$

Y_k and Y_ω represent the dissipation of k and ω due to turbulence and is given by

$$Y_k = \rho \beta^* f_{\beta^*} k \omega;$$

where

$$f_{\beta^*} = \begin{cases} 1, x_k \le 0 \\ \frac{1 + 680 x_k^2}{1 + 400 x_k^2}, x_k > 0 \end{cases}$$

$$x_k = \frac{1}{\omega^3}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}$$

$$\beta^* = \beta_i^*[1 + \varsigma^* F(M_t)]$$

$$\beta_i^* = \beta_\infty^*\left(\frac{4/15 + (\mathrm{Re}_t / R_\beta)^4}{1 + (\mathrm{Re}_t / R_\beta)^4}\right)$$

$\zeta^* = 1.5$, $R_\beta = 8$, $\beta^*_\infty = 0.09$.

$$Y_\omega = \rho \beta f_\beta \omega^2$$

$$f_\beta = \frac{1 + 70 x_\omega}{1 + 80 x_\omega}$$

$$x_\omega = \left|\frac{\Omega_{ij}\Omega_{jk}S_{ki}}{(\beta_\infty^* \omega)^3}\right|$$

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)$$

S_{ki} is the strain rate tensor and

$\beta = \beta_i\,[1 - \frac{\beta_i^*}{\beta_i}\varsigma^* F(M_t)]$ where $F(M_t)$ is the compressibility correction.

D_ω represents the cross-diffusion term, and is defined as

$$D_\omega = 2(1-F_1)\rho\sigma_{\omega,2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}\ ;$$

The model constants used in this study are

$\alpha^*_\infty = 1$, $\alpha_\infty = 0.52$, $\alpha_0 = 1/9$, $\beta^*_\infty = 0.09$, $\beta_i = 0.072$, $R_\beta = 8$, $R_k = 6$, $R_\omega = 2.95$, $\zeta^* = 1.5$, $M_{t0} = 0.25$, $\sigma_k = 2$, $\sigma_\omega = 2$, $\sigma_{k,1} = 1.176$, $\sigma_{\omega,1} = 2.0$, $\sigma_{k,2} = 1.0$, $\sigma_{\omega,2} = 1.168$, $\alpha_1 = 0.31$, $\beta_{i,1} = 0.075$, $\beta_{i,2} = 0.0828$.

INLET CONDITIONS

Gridding was done in the FLUENT® software pre-processor, GAMBIT®. An excellent quality all-quad 2d mesh, with 40k cells, was generated for the 2-D axi-symmetric simulation (fig. 1e) ensuring y+ ~40 that is required for using with standard k-ε model with wall-functions. ~10k cells were distributed in the pre-diffuser region, ~6k cells in the inner and outer annulus passages each, and the rest for the dump region. For use with Shear Stress Transport SST k-ω model, the same mesh (of ~40k cells) was locally refined to 475k cells within the solver, FLUENT® CFD, to achieve the required y+ ~1, for all the walls.

Starting w/ measured profiles

Since the measurements were available only at five points across the passage height, reasonable and consistent assumption for the near-wall treatment needs to be made for center, inner and outer peaked profiler runs. All the results presented in this report for the cases with measured profiles at inlet use a linear approximation from the last available measurement down to the wall. Another reasonable assumption would be to use $1/7^{th}$ power law. But it has to be noted that, since $1/7^{th}$ power law is strictly applicable for a fully developed turbulent flow it will not exactly match the next to the wall measured point at the diffuser inlet. Moreover, the boundary layers along the inner and outer walls are highly dependent on the wake created behind the profiler that varies with the kind of profiler (center/ inner/outer) used, hence the same $1/7^{th}$ power law approximation will not be suitable for all the three profilers the same way. This leaves us with making approximations for a smooth transition to $1/7^{th}$ power law, from each of the measured profile for the three profilers. With these assumptions, minor impact was noted for the predicted static pressure recovery and loss coefficient. Turbulence quantities (k and ε) at the diffuser inlet, is calculated by assigning turbulence intensity, *I* and length scale *l*, as

$$l = 0.07 D_h$$

$$k = \frac{3}{2}(u_{avg} I)^2$$

Turbulence dissipation rate is then calculated using the relationship

$$\varepsilon = C_\mu^{3/4}\frac{k^{3/2}}{l}$$

Starting w/ calculated profiles

Since the bow-tie profiler geometries are 3-dimensional in nature (Fig. 1b), a 3d analysis of each of center, inner and outer peaked profilers were taken up by simulating the test conditions. GAMBIT® software was used for gridding the inlet geometry through the prediffuser. For the through the profiler (TTP) analysis, an all hexahedral mesh comprised of ~512k cells was used, Fig. 1d. A total of 245 grid points were used in the axial direction over a physical length of 9.2" starting from the upstream dump exit to the diffuser inlet. Similarly 20×96 cells were employed at the inlet section (fig. 1c) over a 18^0 sector with a channel height of 0.76". A very fine mesh was employed near the profiler walls with the minimum cell size of 0.0155" and 0.033" in the axial and radial direction respectively. The dimensions of the upstream dump were 8h*5h in the radial and axial direction respectively, where h is the height of the profiler (0.764"). Subsequent studies involving different reasonable assumptions for the plenum upstream of the bell mouth entrance did not make any noticeable impact on the calculated diffuser inlet conditions.

CFD calculated through the inlet-profiler for the prediffuser inlet plane shown in Figures 2 and 3 clearly show the complexity of the flow field approaching the prediffuser. From iso-axial velocity contours and circumferential variation in the axial velocity (Fig 2), contours of total pressure (Fig. 3a) and turbulence kinetic energy (Fig. 3b), circumferential variation of k (Fig. 3c) and ε (Fig. 3d) at the prediffuser inlet, one can clearly question the validity of simplistic assumptions for specifying the inlet conditions based on measurements of

four 5-element total pressure rakes along with wall static pressures for each rake.

This paper starts calculations with circumferentially averaged measured and CFD-calculated inlet profiles. The follow-on paper[7] will have calculations with 2-D inlet profiles to show back-to-back comparison between 1-D and 2-D inlet profiles as well as 3-D simulation of the diffusion system.

COMPARISON

Comparison between data and predictions are made in the following paragraphs for Runs 3, 4 and 5, center-, inner- and outer-peaked diffuser inlet profiles. A detailed comparison is presented for Run #3 to help convey the extent of the analysis conducted in this investigation. Since the details in regard to comparison between data and model predictions for the Runs 4 and 5 are not too different from that of Run #3, only summary tables are presented for the other two runs.

Run #3 Center-Peak Profile

Results are presented in the following paragraphs for C_P and Ω of the combustor inner and outer annuli, pre-diffuser and dome.

C_P of Outer and Inner Passages

Fig. 4 shows typical comparison between C_P data and predictions with the standard k-ε model in conjunction with the use of standard wall function. The calculations are presented with the measured inlet profiles (identified as stdke_wallfunc_meas) and calculated profiles, indicated by stdke_wallfunc_pred. Abscissa represents flow through the outer or inner annuls as appropriate, expressed as fraction of the total flow; y-axis presents C_P. 14 out of 18 measured points were used for the model calculations so that the trend lines for predictions and data can be compared qualitatively. Data and calculations are presented for both the outer and inner passages for stations 3.4 (closer to annulus inlet), and 3.5, toward the end of the annulus.

With the measured inlet profiles, the standard k-ε model over predicts outer and inner passages C_P with considerable error defined as [(Data-Prediction)/Data]. On the average, this most popular current practice approach over-predicts C_P for the two axial stations of the outer passage by 26 and 30%, respectively; see entries in Table 4. On the other hand, model when used with the through-the-profiler calculations (so-called with calculated inlet profiles for mean velocity components, k and ε) over-predicts CP by 6% and 13%, respectively. Similarly for the inner passage, model's over-predictions of 30 and 26% (w/ the measured profiles) reduce to 12% and 10%, respectively. This means, with calculated inlet profiles, the standard k-ε model w/ standard wall function over predicts annuli C_P by ~10% compared to 28% w/ measured profiles; see Table 5. Perhaps, a plausible explanation for this unexpected conclusion is that for most of the engineering-purpose diffuser rigs we don't have sufficiently resolved measurements for the mean quantities; turbulence quantities are rarely measured for these set-ups. One should not expect four 5-element Pt rakes with wall static pressures to pick up representative flow field characteristics. Moreover, it is not easy to measure combustor diffuser inlet profiles in gas turbine engines. We therefore feel that is better to have a process for numerically calculating inlet profiles through a well posed through the inlet simulation than to rely 4-5 Pt rakes to specify inlet conditions.

Figure 5 shows back-to-back comparison between the standard k-ε/wall function and SST low-Reynolds number k-ω models, both w/ measured inlet profiles. The SST model appears to have slight advantage over the k-ε model, the former w/ average over predictions of ~25% compared to ~28% for the k-ε model. However, when compared for the calculated inlet profile, Figure 6, the average over prediction by SST is ~16% compared to ~10% by the k-ε model. This is rather a disappointment considering: 1) the SST calculations are with ~475k cells compared to ~40k cells used by the k-ε model; 2) SST was specifically formulated and validated for flows involving adverse pressure gradients. However when appropriate, we are in favor of using SST for diffuser calculations.

Figure 7 compares the standard k-ε model predictions with the two wall functions (standard and non-equilibrium), measured and calculated inlet profiles. Even though the use of non-equilibrium wall function did not show any advantage over the standard wall function (average over-prediction for the passages being ~11% compared to ~10% w/ the standard wall function), we recommend its use because in future investigations we plan to replace it with wall functions appropriate for flows involving reverse-flow and separation with the hope of getting better correlation between model and data. The built-in advantage of including stream-wise pressure-gradient term in the non-equilibrium wall function is quite evident even though it has not shown any benefit for Run 3.

Ω of Outer and Inner Passages

Figure 8 shows a comparison between the measured and predicted total pressure loss coefficient, Ω with the standard k-ε/wall function starting with measured and calculated inlet profiles. Abscissa represents flow through the outer or inner annuls, expressed as fraction of the total flow; y-axis presents Ω. Average error reduces from ~30% (with measured inlet profiles) to ~13% with the calculated diffuser inlet profiles.

Back to back comparison between the standard k-ε and SST models with the measured inlet profiles, Figure 9 shows that the average error (~30% w/ the former) reduces to ~28%, showing no noticeable advantage of SST over the k-ε model. On the other hand predictions with the calculated diffuser inlet profiles show that average errors of the two models are ~13 and ~18%, indicating the inferior performance of SST model.

Pre-diffuser, Dome C_P, Ω

Unlike the outer and inner passages, the flow split variations don't impact the prediffuser's CP and Ω. Therefore, the comparison between data and model predictions can be shown through a summary Table 4. All the three approaches (SST and standard k-ε models with standard or nonequilibrium wall function) with measured inlet profiles overpredict CP by ~22. The k-ε model w/ calculated inlet values has average error <0.5%; SST model on the other hand over-predicts by 10%. There is no consistent trend predicted for the prediffuser exit total pressure loss coefficient Ω; some of it could possibly be due to the approximation used in calculating experimental value of Ω. However, CFD calculated Ω that fall in the range 0.09 to 0.123 look reasonable.

2-D calculated dome total pressure recovery compare poorly with the data for this particular combustor diffuser set-up that involves high flow through the dome. We could not get 2-D CFD reproduce the data after having simulated the set-up in several reasonable ways. However, 3-D CFD simulation as reported in Reference 7 did considerably better job in correlating with data.

In summary, as given in Tables 4 and 5, 2-D CFD with calculated diffuser inlet flow conditions over-predicted CP of prediffuser, outer and inner passages (stations 3.4 and 3.5) on an average by 0.4%, 5.8%, 13.1%, 12.1% and 10.1%, respectively. The corresponding under-prediction for Ω was 7.1%, 6.1%, 13.7%, 16.5% and 14.9%, respectively.

Comparison between Runs 3, 4 and 5

It is a common practice to document performance of combustor diffusers with different inlet profiles that are created by bow-tie profilers as shown previously in Figure 1b. It is also well recognized that 3-5 multi-element static and total pressure rakes can not do full justice in characterizing the inlet and exit profiles of the prediffuser. For example, Figure 2 shows the predicted diffuser inlet axial velocity contours produced by the three profilers corresponding to Runs 3 (center-peak), 4 (inner-peak) and 5 (outer-peak), respectively. Clearly, circumferentially averaged radial profile from CFD if it agrees with data averaged from 3-5 rakes should be considered coincidental. Consequently, the flowfield development through the prediffuser will be different for each of the six modeling approaches studied in this paper.

Figure 10 for example compares inlet and exit profiles calculated by the six modeling approaches with data. Data can't be explained intuitively. However, model trends are explainable even though the agreement between data and calculations is rather poor, Figure 11. As will become evident from the studies of References 7 through 15, we don't expect models to agree with data from engineering diffuser rigs with bow-tie type profilers The turbulence levels generated by bow-tie profilers are distinctly different from those studied in typical research diffuser set-ups. Nevertheless, combustion designer needs to use CFD for designing the combustor diffusion system, and it is for this reason, we present comparison between data and predictions for Runs 4 and 5, as summarized in Table 5. From this table and tables for Runs 4 and 5 similar to Table 4 for Run 3, we summarize the following findings:

For Run 4, average errors for prediffuser, dome, outer passage (stations 3.4 and 3.5) and inner passage (stations 3.4 and 3.5) CP are 2.1, 25.8, -3.1, -7.9, -15.9 and –11.2%, respectively. Recall, positive error means underprediction and negative values mean overprediction. The corresponding errors in total pressure loss coefficient are: -23.2, -72, 3.3, 8.5, 17.0 and 12.5%.

CP errors for Run 5 are: 11.7, 33.7, 0.98, -2.1, 4.3 and 5.8%. The corresponding errors in total pressure loss coefficient are: -57.6, -230, -1.5, 2.6, -10.4 and –14.6%.

Summary

This is a 1st paper in the planned ten papers to conduct systematic assessment of the available CFD and turbulence modeling capabilities for predicting performance of combustor diffusion systems used in gas turbine engines. As summarized in Table 5, we have conducted 2-D diffusion system simulations with the standard k-ε (with two different wall functions – standard and non-equilibrium) and SST k-Ω turbulence models. 3-D CFD simulation through the three profilers (that give respectively center-, inner- and outer-peak diffuser inlet profiles) clearly showed that the measured inlet profile (with four 5-element total pressure rakes along with wall static pressures) couldn't capture the complexity of the flow field. Consequently, the model predictions starting with measured profiles performed poorly against the simulation with the calculated inlet profiles. 2-D CFD for the diffusion system with high-dome flow predicted the dome pressure recovery poorly.

However, as shown in Reference 7, 3-D CFD simulation gives good agreement.

Prediffuser CP was predicted rather very well (average error <2% which is within the experimental accuracy) for Runs 3 and 4; but the error went up to ~12% for Run #5, outer-peak profile, which we have not been to explain. On the other hand, the average error for the passage recovery is only 2% (under-prediction) for Run #5 compared to approximately ~10% (CP over-predicted) for Runs 3 and 4. 3-D simulation reported in Ref. 7 partially explains this discrepancy.

Figures 12 and 13 show pictorially comparison between data (along x-axis) and 2-D CFD predictions for CP first with the measured profiles, then w/ calculated values for both the standard k-ε and SST k-ω models, followed by the total pressure loss coefficient Ω for Run 3. Similar sets for Runs 4 and 5 follow. With the current standard practice of measured inlet profiles, average overprediction for CP is 27, 23.5 and 12.5%, respectively for Runs 3, 4 and 5. However, these overpredictions reduce to 7.5, 6 and -2.5%, respectively with the calculated inlet profiles. The corresponding errors with SST k-Ω model are 12.5, 17 and 2.5%. The average underpredictions for Ω are 12.5, 0 are –2.5%.

From Figure 14 for the dome total pressure recovery coefficient predicted with the standard k-ε model with calculated inlet velocity is on the average 27.5% lower than data.

Future Diffuser Modeling Plan

In follow-on publications 7 through15 we plan to study the effect of the following on C_P and Ω:

1. 2-D versus 3-D simulations
2. Different turbulence models and near-wall treatments to capture boundary layer losses for diffusers close to or beyond incipient flow separation.
3. Diffusion system design parameters including inlet profiles, pre-diffuser area ratio, L/H, cowling, blockages due to dome swirlers, nozzle stem and tip, airflow splits between the dome, inner and outer annuli of the combustor liner.
4. Multi-passage diffusers validation
5. Engine to rig correlation.

REFERENCES

1. Karki, K.C, Oeschle, V.L., and Mongia, H.C., "A Computational Procedure for Combustor Diffuser Interaction", ASME GT-90-95, published in J. Eng. Gas Turbines and Power, January 1992, Vol. 114, pp 1-7.
2. Agrawal, A.K, Kapat, J.S., and Yang, T.T., 1998, "An Experimental/Computational Study of Airflow in the Combustor-Diffuser System of a Gas Turbine for Power Generation," J. Eng. For Gas Turbines and Power, v120, pp. 24-33, January 1998.
3. Nikjoo, M., and Mongia, H., 1999, "Predictions of Flows with Adverse Pressure Gradients", AIAA Paper 99-2817.
4. Mongia, Hukam C., 2004, "Perspective of Combustion Modeling for Gas Turbine Combustors", AIAA Paper 2004-0156.
5. FLUENT® CFD
6. B.E. Launder, D.B. Spalding, "The numerical computation of turbulent flows", Computer methods in applied mechanics and Engineering, Vol. 3, pp. 269-289 (1974).
7. Mongia, H., Hsiao, G., Burrus, D., Sreedhar, PSVS, and Rao, A., 2004, "Combustor Diffuser Modeling Part II: Inlet Profiles and 3-D Calculations", AIAA Paper 2004-4169.
8. Mongia, H., Hsiao, G., Burrus, D., Habeeb, K., Sreedhar, PSVS, 2004, "Combustor Diffuser Modeling Part III: Validation w/ Typical Separating Single Passage Diffusers Combustor Diffuser Modeling", AIAA Paper 2004-4170.
9. Mongia, H., Hsiao, G., Burrus, D., Sreedhar, PSVS, and Rao, A., 2004, "Combustor Diffuser Modeling Part IV: Effect of Cowling Geometry, Mixer Size and Nozzle Blockage", AIAA Paper 2004-4171.
10. Mongia, H., Hsiao, G., Ravindra, MVV, and Sreedhar, PSVS, 2004, "Combustor Diffuser Modeling Part V: Validation with a Three Passage Diffuser Rig Data", AIAA Paper 2004-4172.
11. Mongia, H., Hsiao, G., Mueller, M., Dai, Z., Ravindra, MVV, Sreedhar, PSVS, and P. Arvind, 2004, "Combustor Diffuser Modeling Part VI: Validation with a Four Passage Diffuser Rig Data", AIAA Paper 2004-4173.
12. Mongia, H., Hsiao, G., Burrus, D., Sreedhar, PSVS, and Rao, A., 2005, "Combustor Diffuser Modeling Part VII: Validation w/ Data of a Single Passage Diffuser Rig Behind Compressor", AIAA Paper 2005-TBD.
13. Mongia, H., Hsiao, G., Burrus, D., Thomsen, D. and Habeeb, K., 2005, "Combustor Diffuser Modeling Part VIII: Single-Passage Diffuser Combustor Rig to Engine Correlation", AIAA Paper 2005-TBD.
14. Mongia, H., Hsiao, G., Burrus, D., Murthy R., Ramesh Kumar, MC, and Naik, P., 2005, "Combustor Diffuser Modeling Part IX: Effect of Dome Flow on Pressure Recoveries", AIAA Paper 2005-TBD.
15. Mongia, Hukam C., 2005, "Perspective of CFD Modeling of Gas Turbine Combustor Diffusers", AIAA Paper 2005-TBD.

Table 1: Typical 3-D Combustor Diffuser Runs

Run #	Comments
3	Center-peaked inlet profile.
4	Inner-peaked inlet profile.
5	Outer-peaked inlet profile.
10	Repeat of Run #3
6	Center-peaked inlet profile w/o cowling
7	Center-peaked inlet profile w/ reduced cowling open area
8	Opposed ring inlet profiler to compare w/ Run 3
9	Center-peaked profile; increased area shorter diffuser
11	Center-peaked profile; increased nozzle dia to 1.9", slightly modified cowling.
12	Run 11 w/ 20% increased nozzle blockage.

Table 2: Summary of prediffuser C_P and σ

Run #	C_P	σ	
		Value	% C_P
3	0.4933	0.0078	1.59%
4	0.5037	0.0112	2.23%
5	0.5665	0.0111	1.97%
10	0.5003	0.0132	2.64%
6	0.5104	0.0048	0.94%
7	0.5067	0.0050	0.99%
8	0.4346	0.0079	1.81%
9	0.4269	0.0132	3.09%
11	0.4980	0.0117	2.34%
12	0.4733	0.0105	2.22%
Avg	0.4914	0.0101	2.08%
σ	0.0398	Pool sigma	
%	8.09%		

Table 3: Summary of prediffuser Ω and σ

Run #	Ω	σ	
		Value	% Ω
3	0.1397	0.0076	5.47%
4	0.1295	0.0110	8.46%
5	0.0681	0.0109	15.99%
10	0.1327	0.0129	9.71%
6	0.1229	0.0047	3.80%
7	0.1270	0.0049	3.87%
8	0.1971	0.0077	3.90%
9	0.2042	0.0129	6.31%
11	0.1349	0.0114	8.44%
12	0.1601	0.0102	6.38%
Avg	0.1416	0.0098	8.05%
σ	0.0388	Pool sigma	
%	27.43%		

Table 4: Comparison between average (for the entire range of Run #3, Center-Peak Profile) measured and predicted CP and total pressure loss coefficient w/ the six modeling/inlet condition approaches.

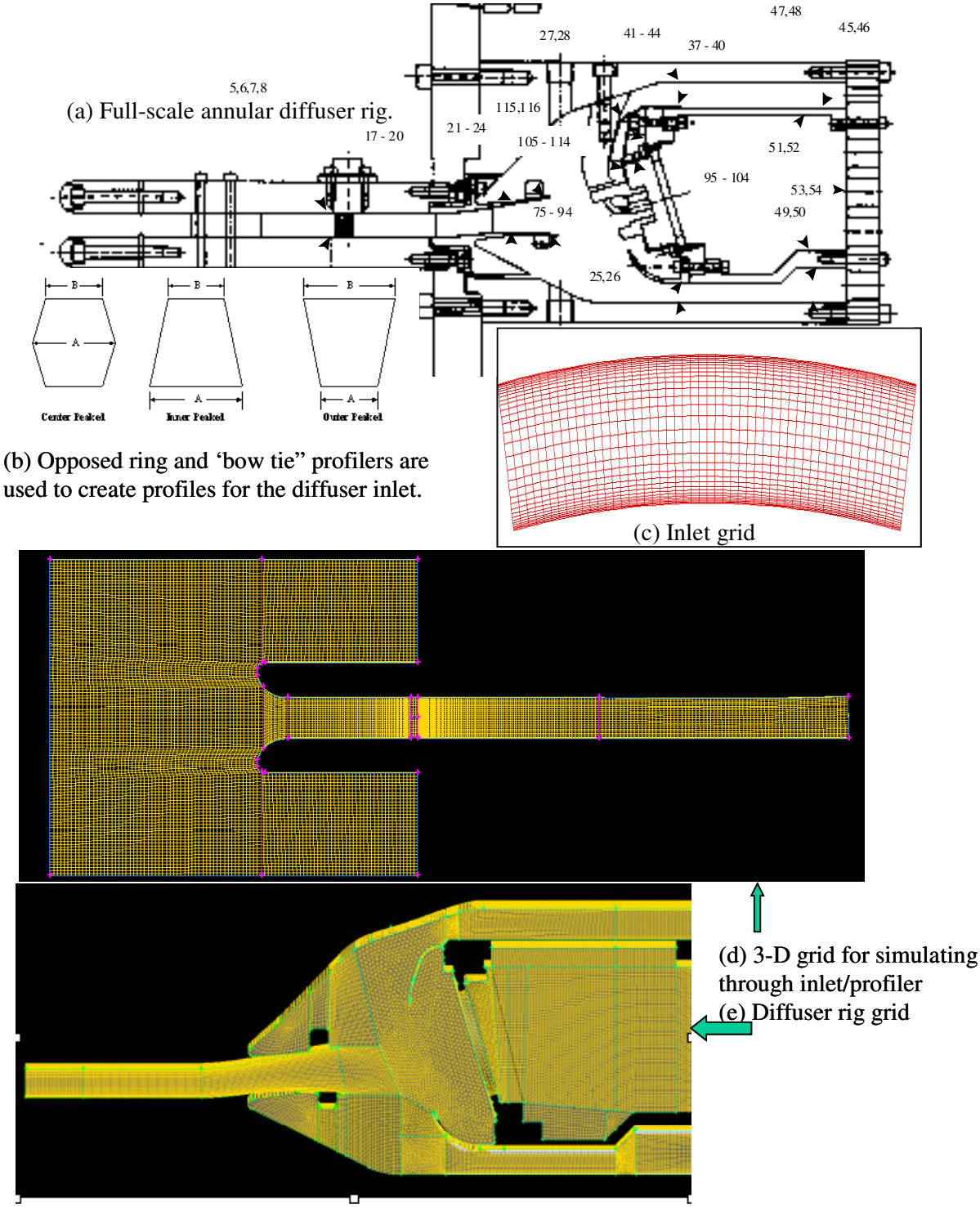
		Prediffuser Exit						Combustor Dome					
Run # 3 Center-Peak Profile		Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff	Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff
Std k-ε, Std wall func	Avg	0.493	0.603	-21.63%	0.140	0.117	14.77%	0.840	0.712	15.97%	0.160	0.288	-95.88%
w/ measured inlet profiles	σ	0.008	0.006	1.09%	0.008	0.006	3.90%	0.037	0.025	2.33%	0.037	0.025	29.48%
Approach #1	σ%/Max	1.59%	1.03%	-24.58%	5.47%	5.33%	24.32%	4.39%	3.45%	18.04%	23.04%	8.52%	-32.97%
Std k-ε, Std wall func	Avg	0.493	0.494	**0.36%**	0.140	0.127	**7.05%**	0.840	0.641	**24.40%**	0.160	0.359	-144.28%
w/ calculated inlet profiles	σ	0.008	0.008	**1.43%**	0.008	0.008	**5.17%**	0.037	0.030	**2.30%**	0.037	0.030	37.45%
Approach #2	σ%/Max	1.59%	1.67%	**-3.08%**	5.47%	6.47%	**19.02%**	4.39%	4.72%	**27.19%**	23.04%	8.41%	-65.61%
SST k-ω model	Avg	0.493	0.596	-20.18%	0.140	0.090	34.51%	0.840	0.677	20.12%	0.160	0.323	-118.61%
w/ measured inlet profiles	σ	0.008	0.002	0.61%	0.008	0.002	1.54%	0.037	0.034	2.42%	0.037	0.034	30.21%
Approach #3	σ%/Max	1.59%	0.33%	-21.43%	5.47%	2.48%	37.58%	4.39%	5.08%	22.81%	23.04%	10.65%	-51.32%
SST k-ω model	Avg	0.493	0.541	-9.03%	0.140	0.107	21.81%	0.840	0.628	25.92%	0.160	0.372	-151.89%
w/ calculated inlet profiles	σ	0.008	0.002	0.51%	0.008	0.002	1.32%	0.037	0.038	2.60%	0.037	0.038	35.28%
Approach #4	σ%/Max	1.59%	0.37%	-10.07%	5.47%	1.87%	24.00%	4.39%	6.06%	29.83%	23.04%	10.23%	-74.46%
Std k-ε	Avg	0.493	0.605	-22.00%	0.140	0.112	18.27%	0.840	0.709	16.30%	0.160	0.291	-97.65%
Nonequil wall function	σ	0.008	0.006	1.07%	0.008	0.006	3.83%	0.037	0.025	2.24%	0.037	0.025	29.54%
w/ measured inlet profiles	σ%/Max	1.59%	1.01%	-24.94%	5.47%	5.48%	27.70%	4.39%	3.54%	18.31%	23.04%	8.62%	-35.45%
Std k-ε	Avg	0.493	0.497	-0.25%	0.140	0.123	10.45%	0.840	0.636	24.98%	0.160	0.364	-147.50%
Nonequil wall function	σ	0.008	0.008	1.40%	0.008	0.008	5.07%	0.037	0.031	2.20%	0.037	0.031	37.56%
w/ calculated inlet profiles	σ%/Max	1.59%	1.62%	-3.68%	5.47%	6.58%	22.28%	4.39%	4.86%	27.63%	23.04%	8.49%	-69.27%

Approach		Ouer Passage Station 3.4						Inner Passage Station 3.4					
#		Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff	Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff
1	Avg	0.503	0.626	-26.37%	0.481	0.360	25.91%	0.503	0.671	-29.52%	0.377	0.252	32.66%
	σ	0.025	0.024	2.69%	0.015	0.017	2.47%	0.083	0.042	9.30%	0.025	0.010	2.94%
	σ%/Max	4.87%	3.83%	-30.27%	3.10%	4.67%	29.03%	16.53%	6.29%	-52.80%	6.59%	4.12%	37.58%
2	Avg	0.503	0.525	**-5.77%**	0.481	0.456	**6.11%**	0.503	0.583	**-12.06%**	0.377	0.312	**16.53%**
	σ	0.025	0.044	**5.50%**	0.015	0.035	**5.62%**	0.083	0.057	**3.81%**	0.025	0.014	**3.08%**
	σ%/Max	4.87%	8.31%	**-10.45%**	3.10%	7.59%	**11.54%**	16.53%	9.84%	**-21.51%**	6.59%	4.37%	**22.15%**
3	Avg	0.503	0.611	-23.42%	0.481	0.373	23.12%	0.503	0.659	-27.33%	0.377	0.254	32.15%
	σ	0.025	0.015	3.92%	0.015	0.009	2.35%	0.083	0.040	10.19%	0.025	0.032	7.44%
	σ%/Max	4.87%	2.45%	-31.59%	3.10%	2.34%	27.85%	16.53%	6.08%	-47.49%	6.59%	12.70%	37.88%
4	Avg	0.503	0.559	-12.82%	0.481	0.423	12.70%	0.503	0.611	-17.81%	0.377	0.292	22.07%
	σ	0.025	0.017	3.55%	0.015	0.010	2.79%	0.083	0.046	8.18%	0.025	0.036	8.37%
	σ%/Max	4.87%	3.00%	-19.82%	3.10%	2.40%	17.72%	16.53%	7.57%	-30.92%	6.59%	12.40%	29.02%
5	Avg	0.503	0.630	-27.04%	0.481	0.356	26.65%	0.503	0.672	-29.67%	0.377	0.250	33.11%
	σ	0.025	0.025	2.48%	0.015	0.018	2.58%	0.083	0.043	9.18%	0.025	0.010	2.90%
	σ%/Max	4.87%	4.04%	-30.42%	3.10%	5.08%	29.97%	16.53%	6.39%	-52.72%	6.59%	4.05%	38.02%
6	Avg	0.503	0.535	-7.81%	0.481	0.446	8.13%	0.503	0.587	-12.83%	0.377	0.308	17.73%
	σ	0.025	0.034	3.06%	0.015	0.024	3.35%	0.083	0.059	3.94%	0.025	0.013	3.55%
	σ%/Max	4.87%	6.32%	-11.10%	3.10%	5.37%	12.19%	16.53%	10.04%	-21.27%	6.59%	4.10%	24.09%

Approach		Ouer Passage Station 3.5						Inner Passage Station 3.5					
#		Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff	Cp - DATA	Cp - CFD	% Diff	Ω - DATA	Ω-CFD	% Diff
1	Avg	0.510	0.654	-29.96%	0.474	0.332	30.46%	0.518	0.672	-26.18%	0.363	0.251	30.59%
	σ	0.021	0.012	4.14%	0.012	0.007	2.36%	0.073	0.044	4.77%	0.032	0.010	3.73%
	σ%/Max	4.15%	1.76%	-36.13%	2.55%	2.00%	34.30%	14.10%	6.50%	-32.65%	8.72%	3.94%	37.57%
2	Avg	0.510	0.569	**-13.07%**	0.474	0.411	**13.70%**	0.518	0.588	**-10.05%**	0.363	0.307	**14.93%**
	σ	0.021	0.016	**3.57%**	0.012	0.009	**3.04%**	0.073	0.061	**3.81%**	0.032	0.012	**4.83%**
	σ%/Max	4.15%	2.74%	**-19.02%**	2.55%	2.17%	**18.75%**	14.10%	10.34%	**-17.92%**	8.72%	4.02%	**24.03%**
3	Avg	0.510	0.635	-26.16%	0.474	0.349	26.79%	0.518	0.663	-24.54%	0.363	0.250	31.00%
	σ	0.021	0.008	4.11%	0.012	0.005	2.43%	0.073	0.047	4.81%	0.032	0.021	4.75%
	σ%/Max	4.15%	1.32%	-32.56%	2.55%	1.52%	31.15%	14.10%	7.13%	-31.09%	8.72%	8.25%	38.19%
4	Avg	0.510	0.586	-16.47%	0.474	0.396	16.96%	0.518	0.615	-15.39%	0.363	0.287	20.63%
	σ	0.021	0.009	3.94%	0.012	0.007	3.00%	0.073	0.054	4.40%	0.032	0.023	5.71%
	σ%/Max	4.15%	1.56%	-22.87%	2.55%	1.71%	22.20%	14.10%	8.84%	-23.11%	8.72%	7.99%	29.74%
5	Avg	0.510	0.659	-30.90%	0.474	0.327	31.50%	0.518	0.672	-26.19%	0.363	0.250	30.86%
	σ	0.021	0.012	4.04%	0.012	0.007	2.25%	0.073	0.044	4.64%	0.032	0.009	3.72%
	σ%/Max	4.15%	1.83%	-36.89%	2.55%	2.04%	34.96%	14.10%	6.62%	-32.45%	8.72%	3.74%	37.80%
6	Avg	0.510	0.575	-14.12%	0.474	0.406	14.84%	0.518	0.587	-9.90%	0.363	0.308	14.85%
	σ	0.021	0.015	3.40%	0.012	0.008	2.82%	0.073	0.062	3.84%	0.032	0.011	4.85%
	σ%/Max	4.15%	2.70%	-19.84%	2.55%	2.03%	19.58%	14.10%	10.50%	-17.72%	8.72%	3.71%	23.86%

Table 5: Comparison between average (for the entire range of Runs 3, 4 and 5) measured and predicted CP and total pressure loss coefficient w/ the six modeling/inlet condition approaches.

	Run #3	Run #4	Run #5	Avg	Run #3	Run #4	Run #5	Avg.
Prediffuser Exit Data	0.493	0.504	0.566	14.0%	0.140	0.130	0.0681	63.7%
$(Max\text{-}Min)_{Dome}$ /Avg Prediffuser CP	27.3%	24.4%	24.8%	25.5%				
$(Max\text{-}Min)_{OP}$/Avg Prediffuser CP	17.8%	14.7%	14.4%	15.6%	10.6%	7.7%	8.4%	8.9%
$(Max\text{-}Min)_{IP}$/Avg Prediffuser CP	64.5%	67.4%	61.9%	64.6%	27.9%	32.2%	28.9%	29.7%
Modeling & Inlet Conditions	Avg Error Prediffuser CP				Avg Error Prediffuser Ω			
measured profile_ske with std wall func	-21.63%	-19.87%	-6.19%	17.33%	14.77%	8.54%	-83.37%	49.13%
calculated profile_ske with std wall func	**0.36%**	**2.07%**	**11.69%**	**6.86%**	**7.05%**	**-23.19%**	**-57.55%**	**36.05%**
measured profile_sst k-w	-20.18%	-18.82%	-4.58%	16.15%	34.51%	30.48%	-82.58%	54.59%
calculated profile_sst k-w	-9.03%	-8.98%	2.75%	7.52%	21.81%	4.34%	-95.39%	56.55%
measured profile_ske with NE wall func	-22.00%	-20.10%	-7.03%	17.68%	18.27%	12.16%	-28.55%	20.79%
calculated profile_ske with NEwall func	-0.25%	2.24%	11.04%	6.51%	10.45%	-22.96%	-48.67%	31.65%
Modeling & Inlet Conditions	Avg Error Dome CP				Avg Error Dome Ω			
measured profile_ske with std wall func	15.97%	16.51%	23.92%	19.15%	-95.88%	-4.51%	-160.5%	108.0%
calculated profile_ske with std wall func	**24.40%**	**25.78%**	**33.73%**	**28.27%**	**-144.3%**	**-71.97%**	**-230.4%**	**162.4%**
measured profile_sst k-w	20.12%	20.90%	28.04%	23.30%	-118.6%	-28.80%	-223.0%	146.8%
calculated profile_sst k-w	25.92%	25.41%	30.98%	27.55%	-151.9%	-47.22%	-272.5%	182.2%
measured profile_ske with NE wall func	16.30%	17.04%	24.83%	19.77%	-97.65%	-4.28%	-155.8%	106.2%
calculated profile_ske with NEwall func	24.98%	26.71%	32.14%	28.11%	-147.5%	-156.6%	-517.7%	323.7%
Modeling & Inlet Conditions	Avg Error Passage CP				Avg Error Passage Ω			
measured profile_ske with std wall func	-28.01%	-26.70%	-11.33%	23.28%	29.91%	27.15%	13.79%	24.64%
calculated profile_ske with std wall func	**-10.24%**	**-9.53%**	**2.26%**	**8.18%**	**12.82%**	**10.31%**	**-5.99%**	**10.11%**
measured profile_sst k-w	-25.36%	-24.08%	-7.67%	20.67%	28.26%	25.33%	9.11%	22.54%
calculated profile_sst k-w	-15.62%	-16.52%	-1.17%	13.14%	18.09%	17.98%	-1.47%	14.75%
measured profile_ske with NE wall func	-28.45%	-27.50%	-13.01%	24.05%	30.53%	28.02%	16.68%	25.79%
calculated profile_ske with NEwall func	-11.17%	-10.11%	2.34%	8.80%	13.89%	10.92%	-6.20%	10.81%



(a) Full-scale annular diffuser rig.

(b) Opposed ring and 'bow tie" profilers are used to create profiles for the diffuser inlet.

(c) Inlet grid

(d) 3-D grid for simulating through inlet/profiler

(e) Diffuser rig grid

Figure 1: Diffuser rig schematic and typical grid used for inlet and diffuser calculations.

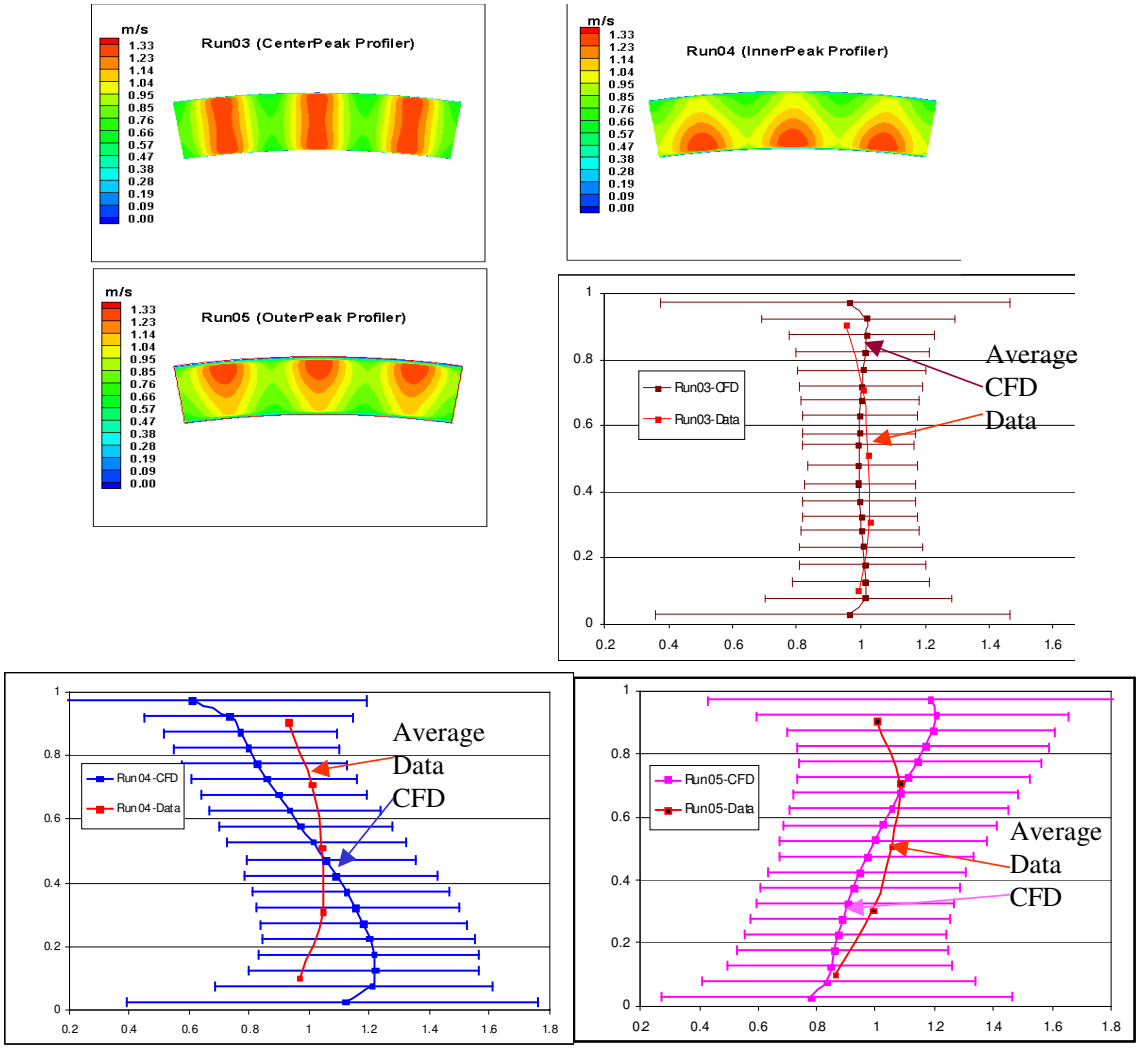
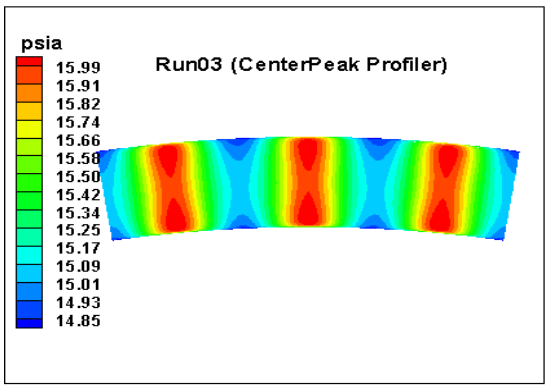
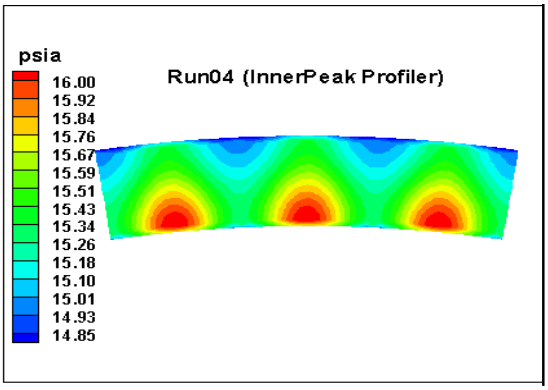


Figure 2: Calculated velocity contours for Runs 3, 4and 5, normalized with average velocity. Diffuser inlet velocity profiles.





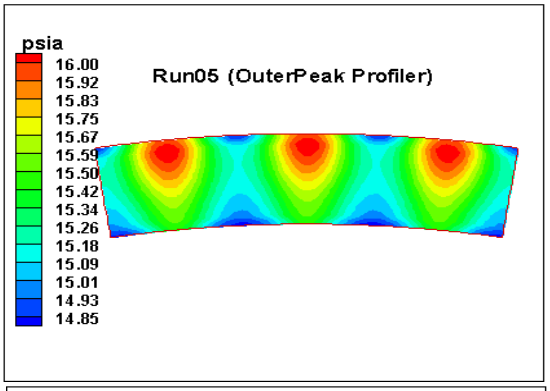
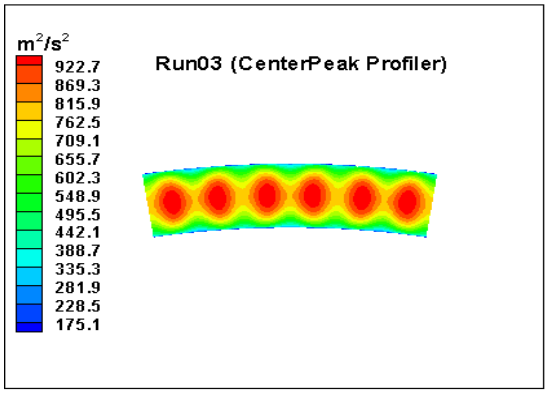
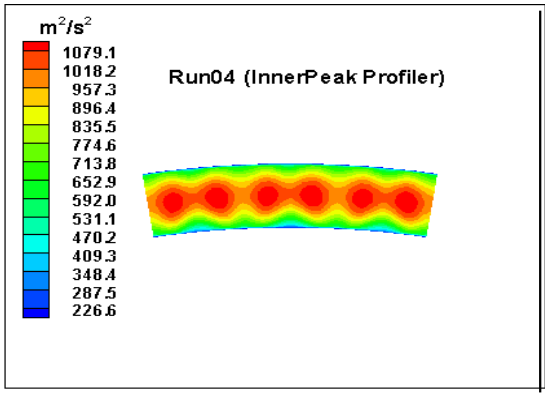


Figure 3a : Calculated 3-dimensional contours of P_t at the diffuser inlet.





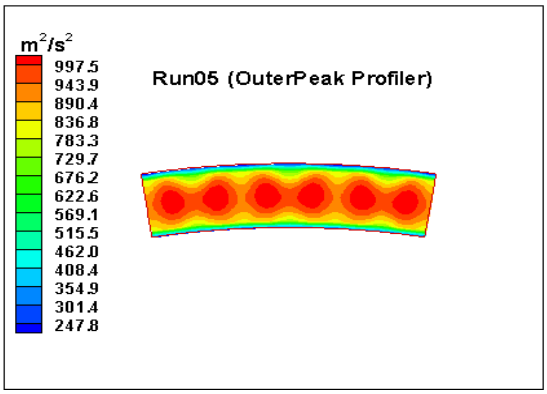
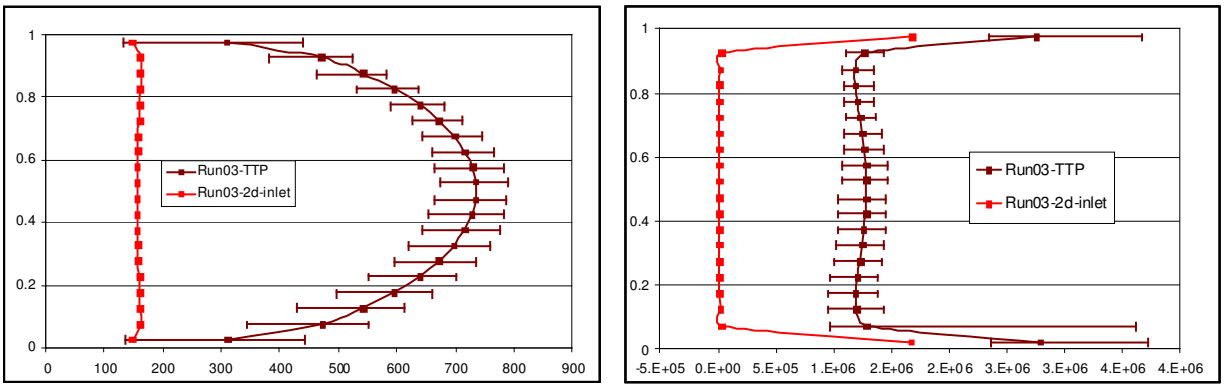
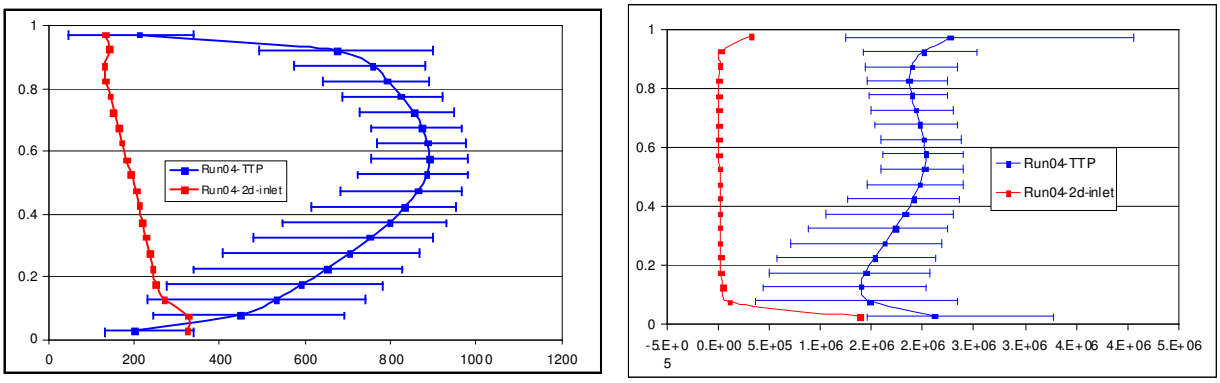


Figure 3b : Calculated 3-dimensional contours of turbulent kinetic energy, k at the diffuser inlet.

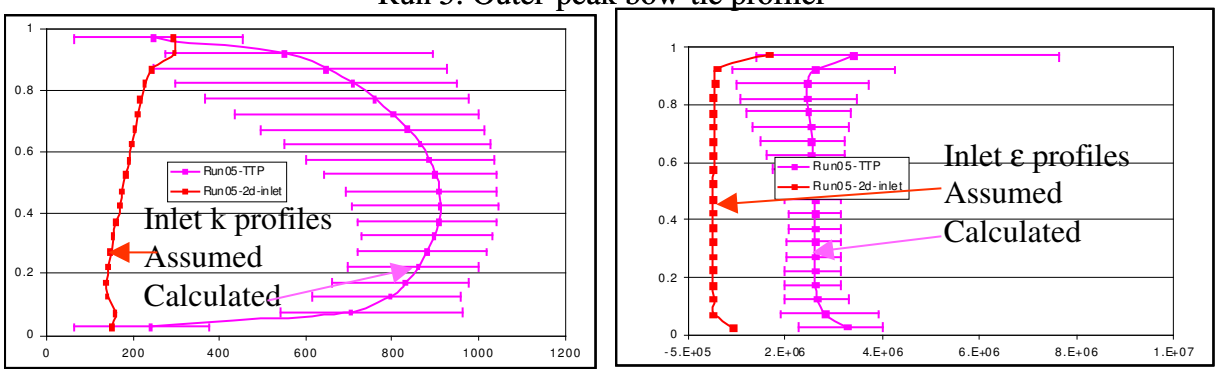
Run 3: Center-peak bow-tie profiler



Run 4: Inner-peak bow-tie profiler

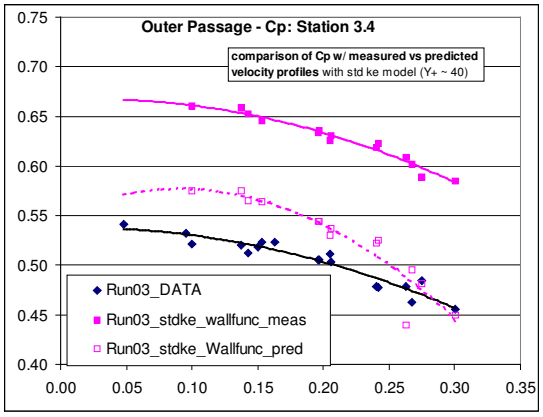


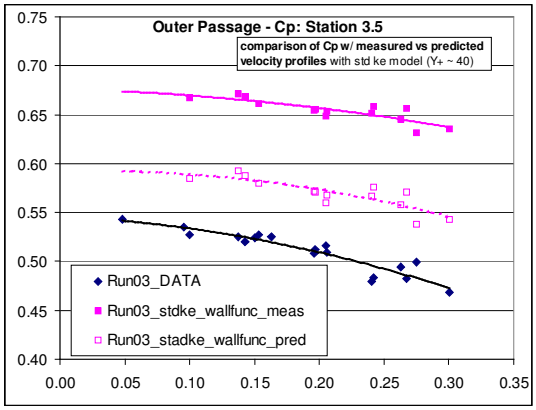
Run 5: Outer-peak bow-tie profiler



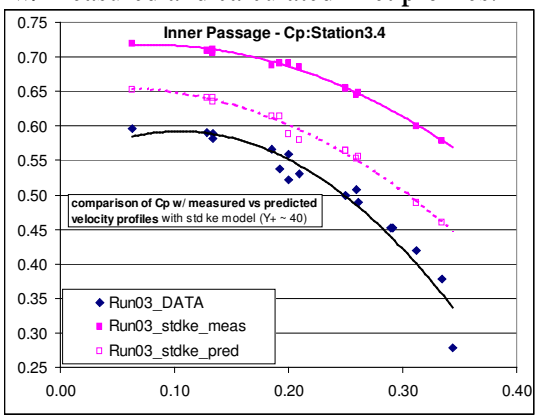
(3c) Diffuser inlet k profiles (3d) Diffuser inlet ε profiles

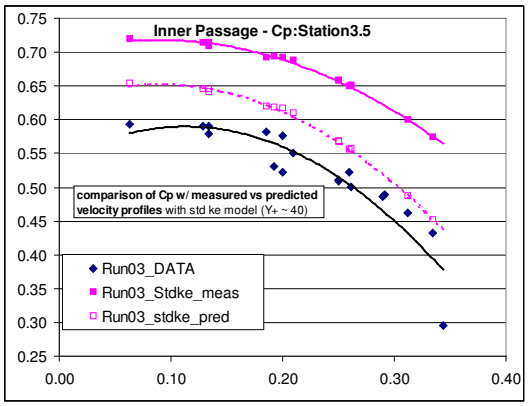
Figure 3: Diffuser inlet conditions for Pt, k and ε for diffuser modeling with measured and CFD-based through the profiler calculations





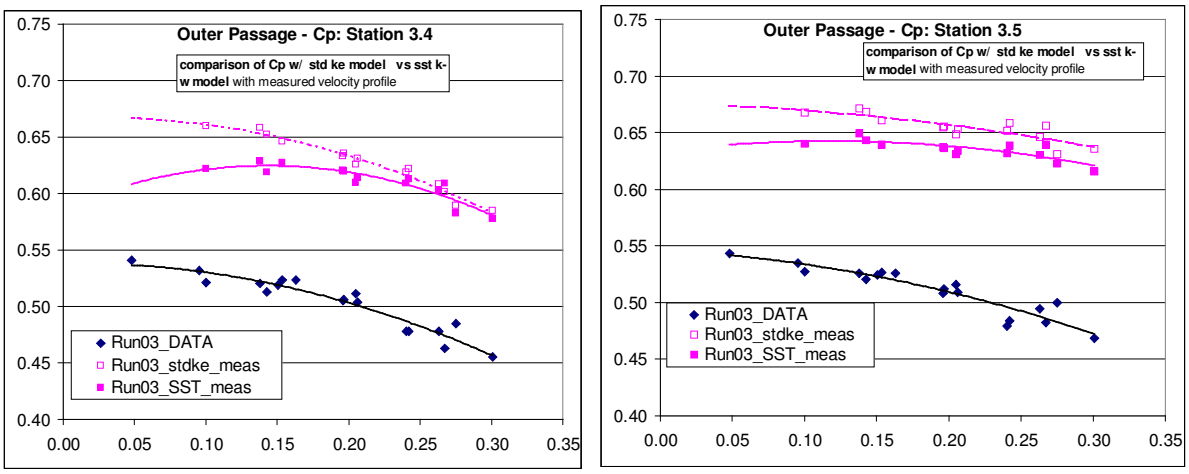
Run # 3 Outer passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions; w/ measured and calculated inlet profiles.



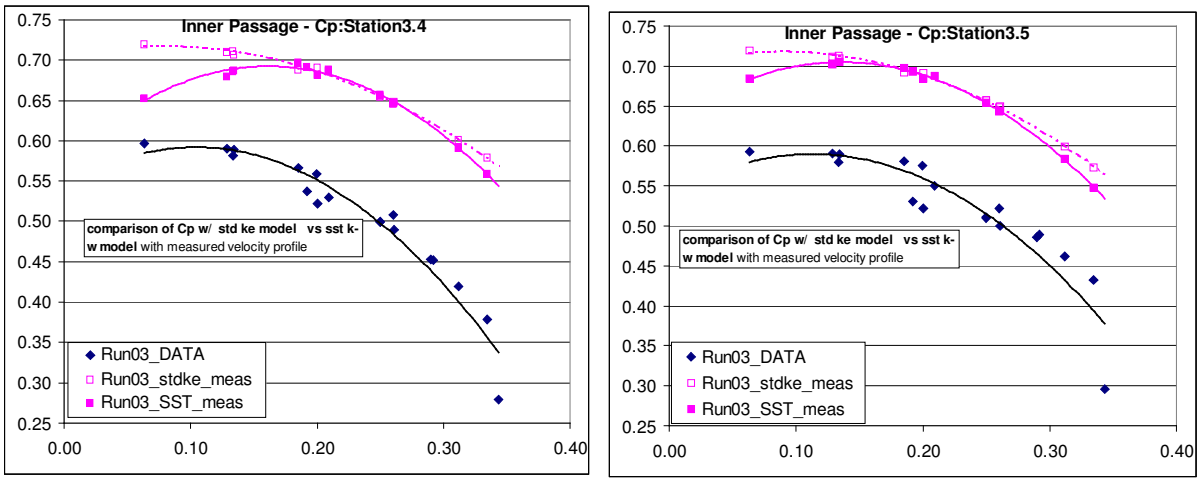


Run # 3 Inner passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions; w/ measured and calculated inlet profiles.

Figure 4: Run 3 passage CP average error between data and predictions ((Data-CFD)/Data) reduces from -28% (overprediction) to -10% by using calculated diffuser inlet profiles.

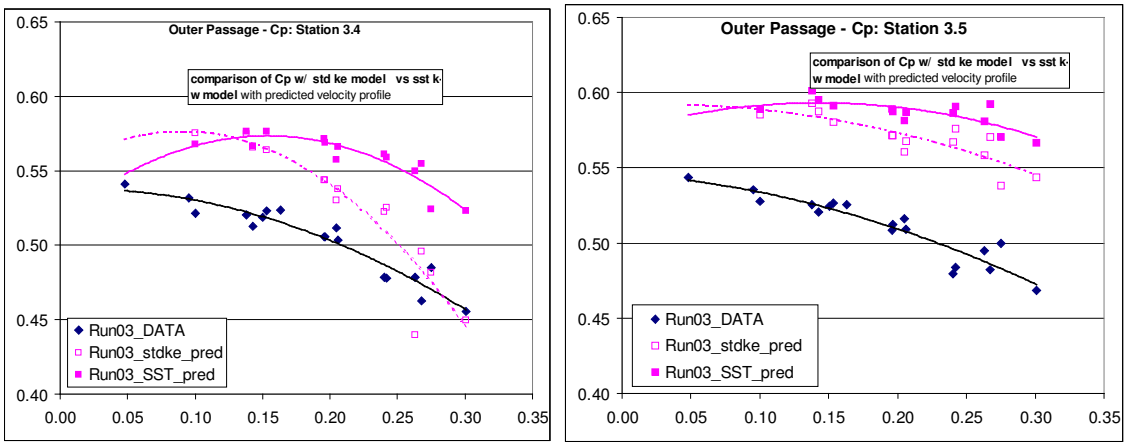


Run # 3 Outer passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions and SST k–ω models; w/ measured inlet profiles.

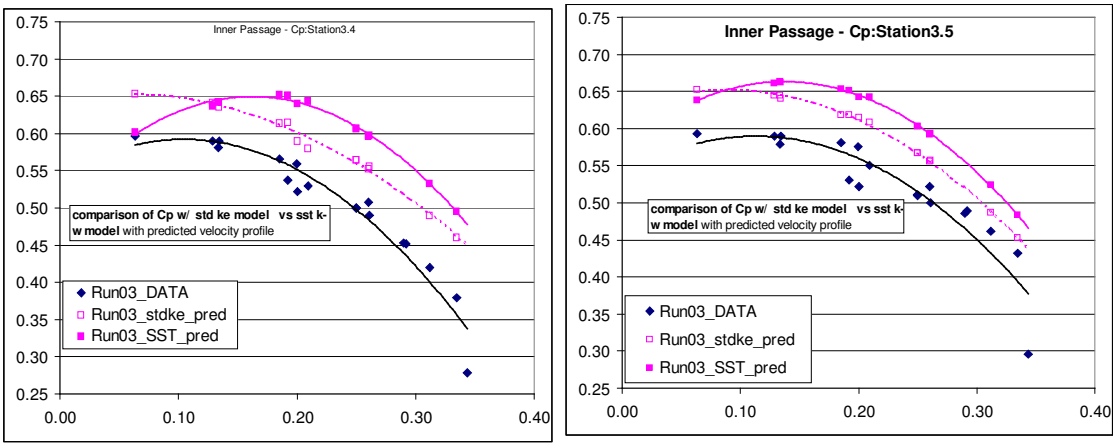


Run # 3 Inner passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions and SST k–ω models; w/ measured inlet profiles.

Figure 5: SST k-ω model slightly better (average error –25%) than the standard k-ε model (-28%) with standard wall functions when calculations performed w/ measured inlet velocity profiles

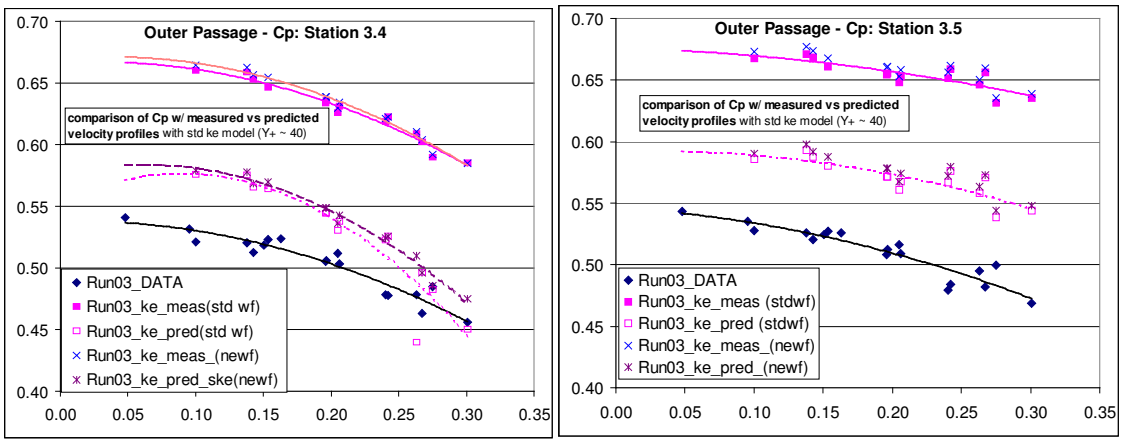


Run # 3 Outer passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions and SST k–ω models; w/ calculated inlet profiles.

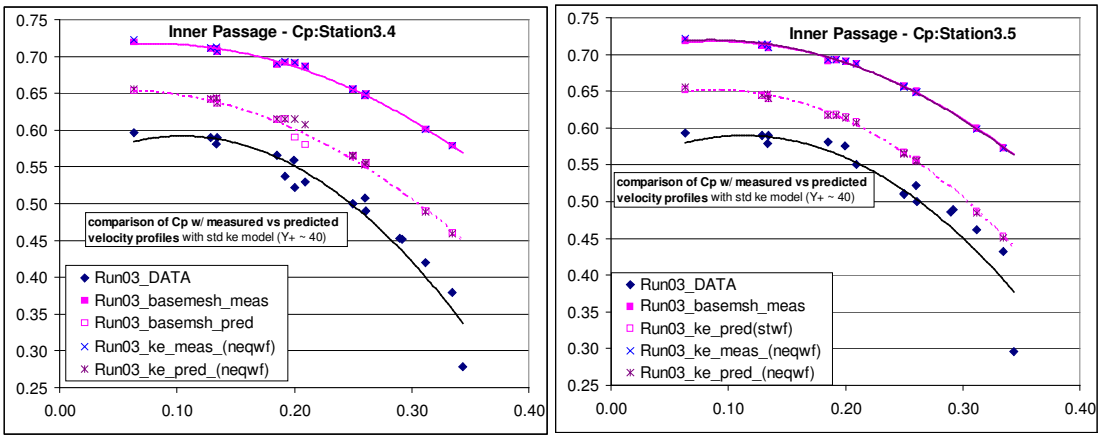


Run # 3 Inner passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, wall functions and SST k–ω models; w/ calculated inlet profiles.

Figure 6: SST k-ω model (avg. error -16%) inferior to the standard k-ε model (-10%) with standard wall functions starting w/ calculated inlet profiles of velocity, k and ε.

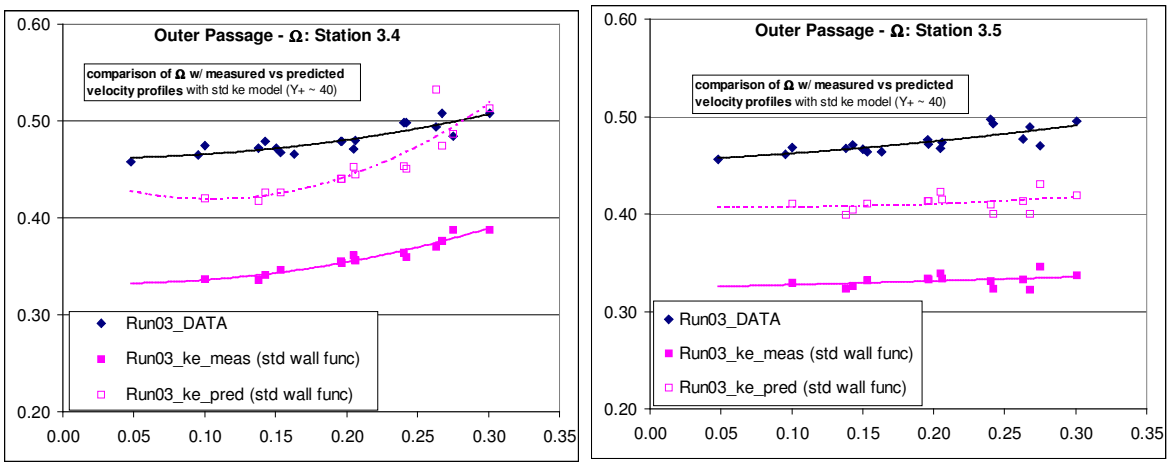


Run # 3 Outer passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, standard and non-equilibrium wall functions; w/ measured and calculated inlet profiles.

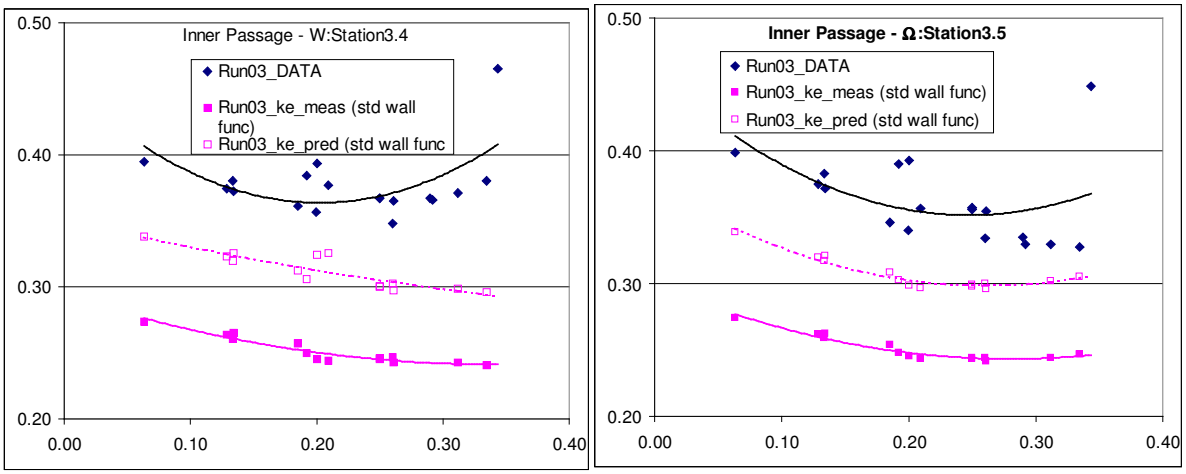


Run # 3 Inner passage stations 3.4 & 3.5 CP data and predictions w/ standard k-ε, standard and non-equilibrium wall functions; w/ measured and calculated inlet profiles.

Figure 7: Non-equilibrium wall functions (average error 11%) don't show any advantage over the standard wall functions (10%). The former should be preferred or replaced by wall functions appropriate for diffusers flows especially when close to incipient separation.

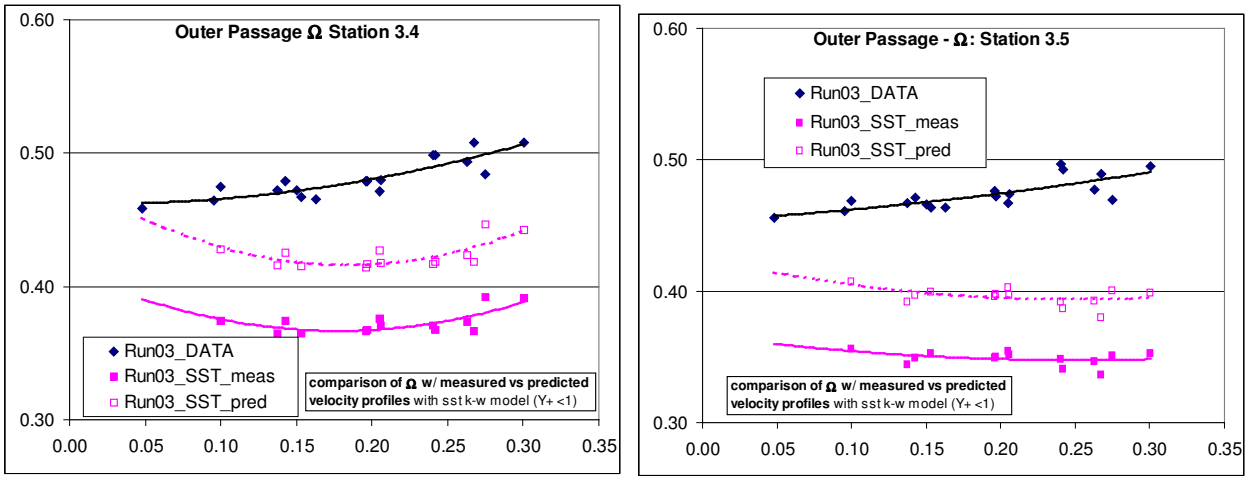


Run # 3 Outer passage stations 3.4 & 3.5 Ω data and predictions w/ the k-ε model; w/ measured and calculated inlet profiles.

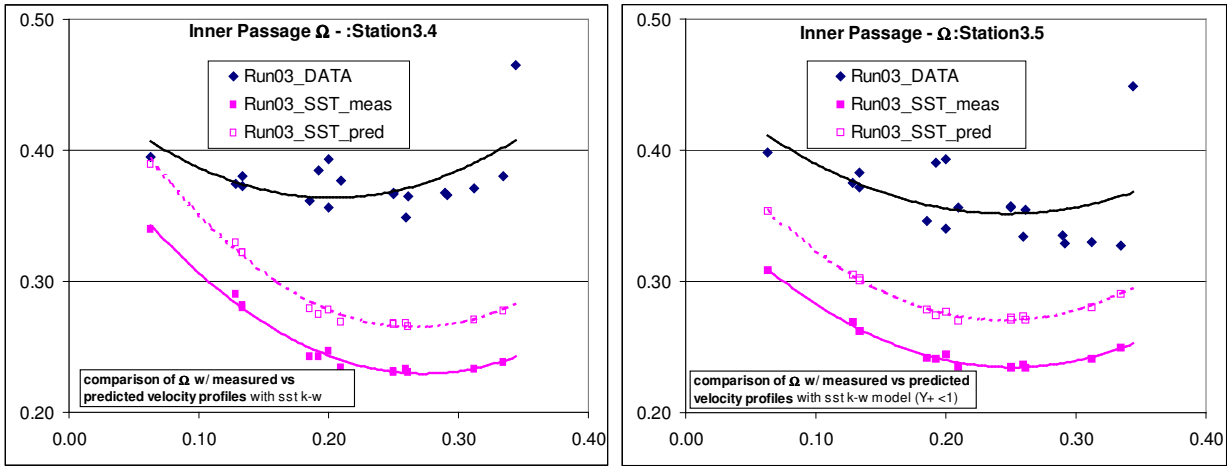


Run # 3 Inner passage stations 3.4 & 3.5 Ω data and predictions w/ the k-ε model; w/ measured and calculated inlet profiles.

Figure 8: Standard k-ε model correlations w/ calculated inlet profiles are better than those w/ measured velocity profiles; average Ω under-prediction being ~30% versus ~13% .

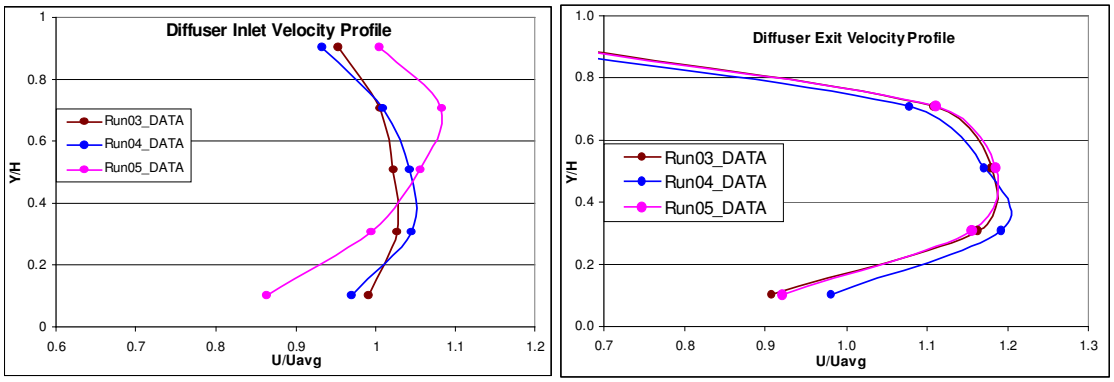


Run # 3 Outer passage stations 3.4 & 3.5 Ω data and predictions w/ SST k-ω, model; w/ measured and calculated inlet profiles.

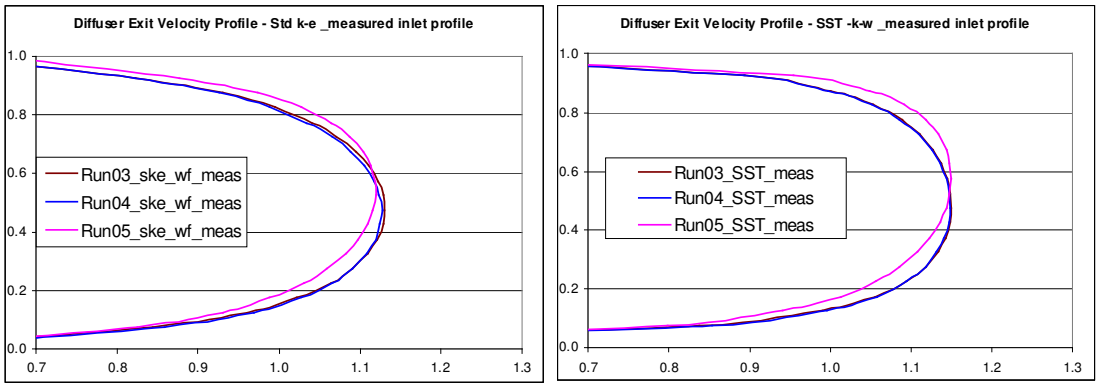


Run # 3 Inner passage stations 3.4 & 3.5 Ω data and predictions w/ SST k-ω, model; w/ measured and calculated inlet profiles.

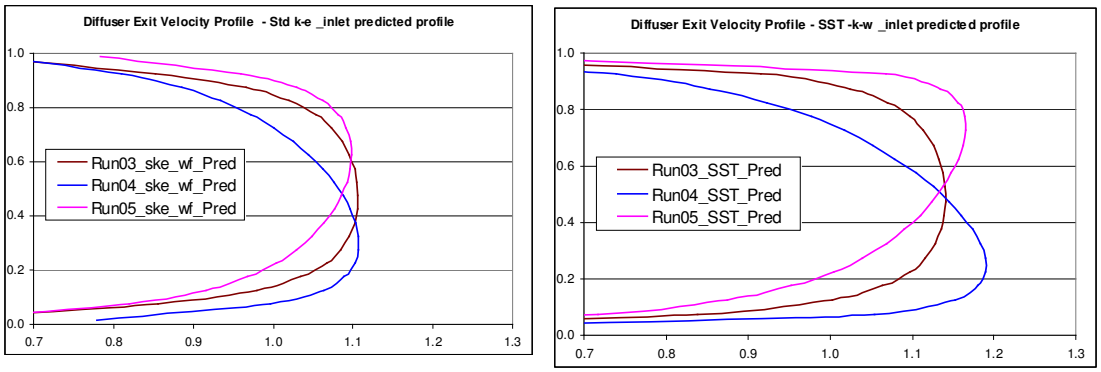
Figure 9: SST k-ω model correlations w/ calculated inlet profiles are better than those w/ measured velocity profiles; average Ω under-prediction being ~28% versus ~18% .



(a) Measured prediffuser inlet and exit velocity profiles



(b) Standard k-ε and SST k-ω models predicted diffuser exit profiles starting w/ measured inlet profiles



(c) Standard k-ε and SST k-ω models predicted diffuser exit profiles starting w/ calculated inlet profiles.

Figure 10: Measured and calculated prediffuser inlet and exit profiles.

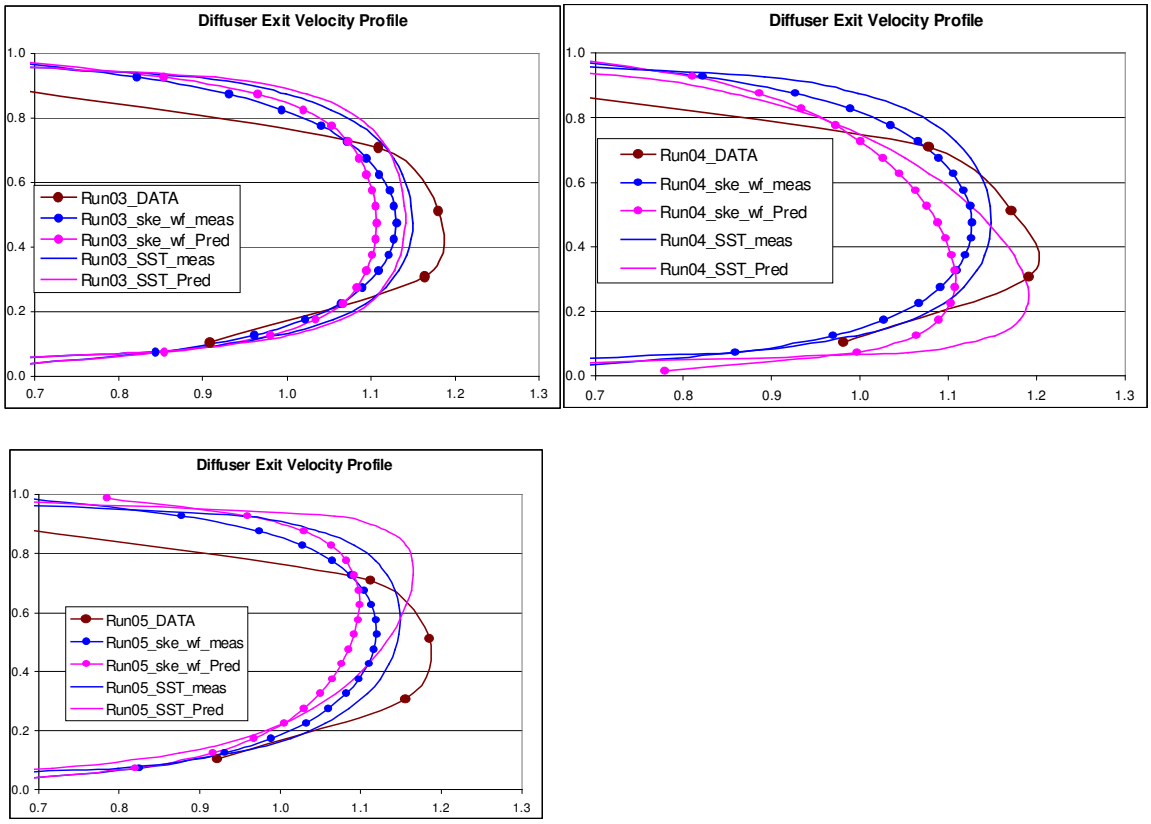


Figure 11: Comparison between measured and calculated prediffuser exit plane velocity profiles for Runs 3, 4 and 5.

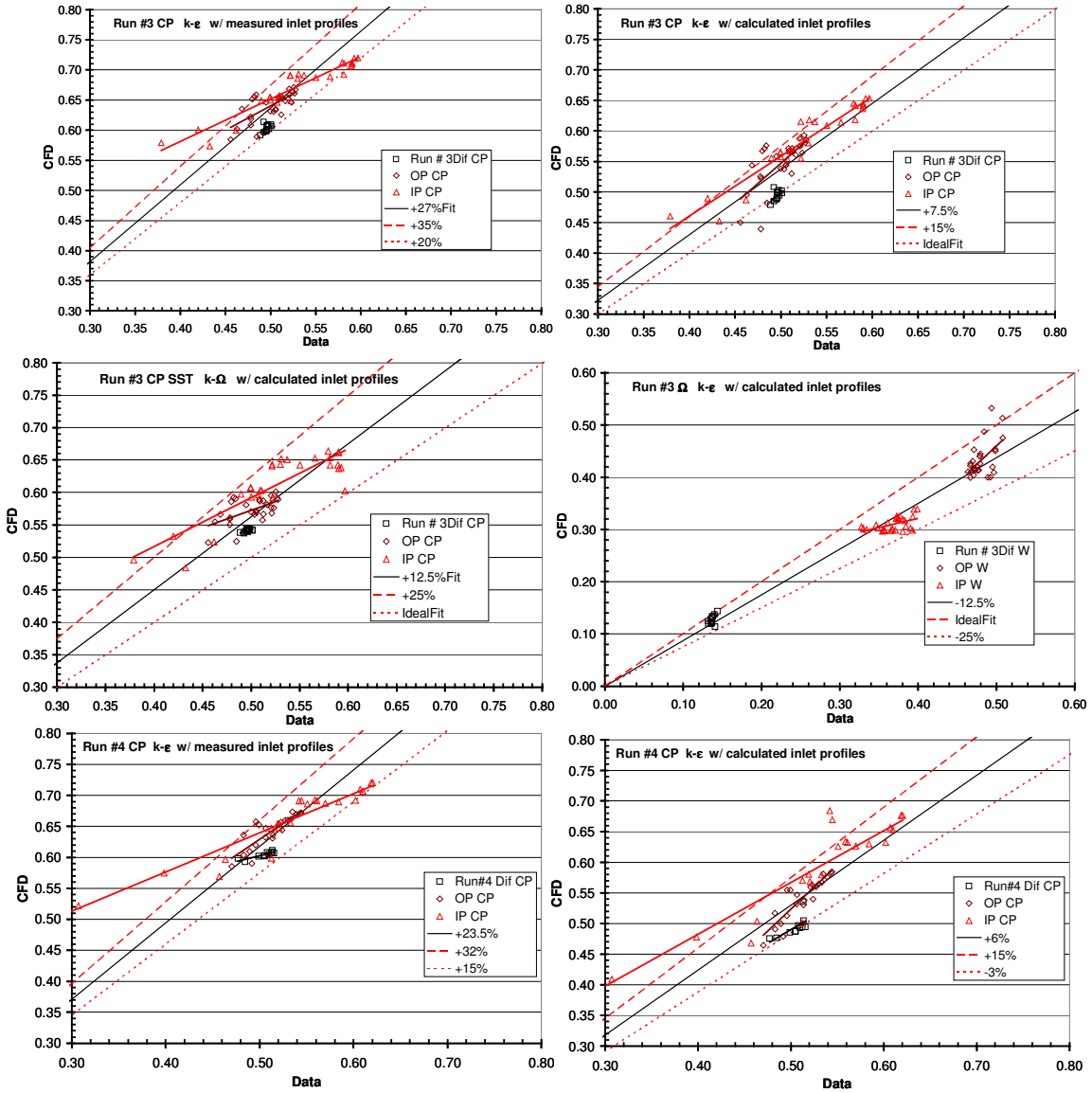


Figure 12: Data versus 2-D CFD results for Runs 3 (center-peak), 4 (inner-peak) and 5 (outer-peak profiler). Top and middle figures: Run 3 CP results measured and calculated diffuser inlet profiles w/ k-ε and standard wall function; CP w/ SST k-Ω and total pressure loss coefficient Ω w/ k-ε model.
Bottom 2 figures: Run 4 CP results measured and calculated diffuser inlet profiles w/ k-ε and standard wall

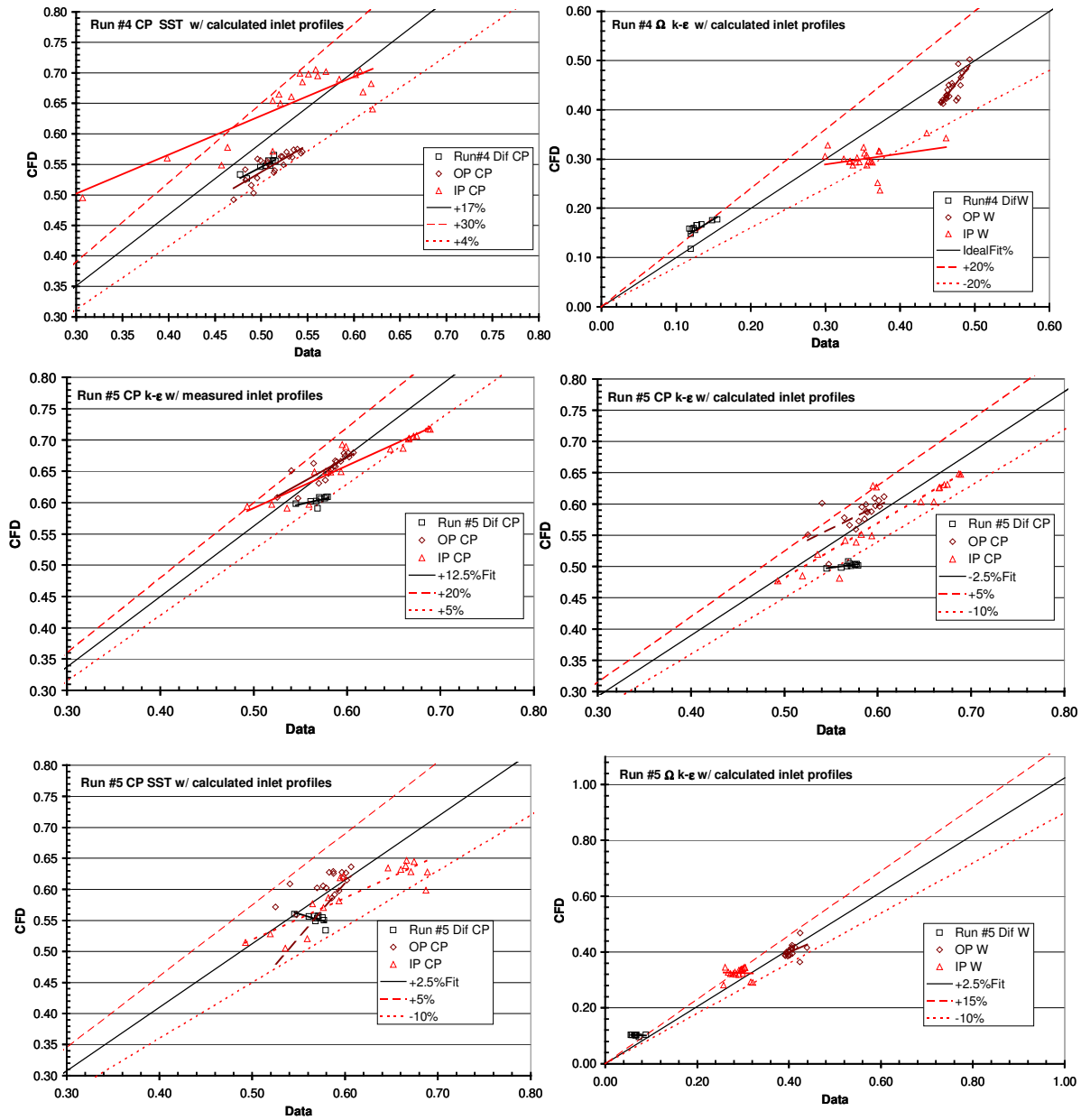


Figure 13: Data versus 2-D CFD results for Runs 3 (center-peak), 4 (inner-peak) and 5 (outer-peak profiler).
Top 2 figures: Run 4 CP w/ SST k-Ω and total pressure loss coefficient Ω w/ k-ε model.
Middle and bottom figures: Run 5 CP results measured and calculated diffuser inlet profiles w/ k-ε and standard wall function; CP w/ SST k-Ω and total pressure loss coefficient Ω w/ k-ε model.

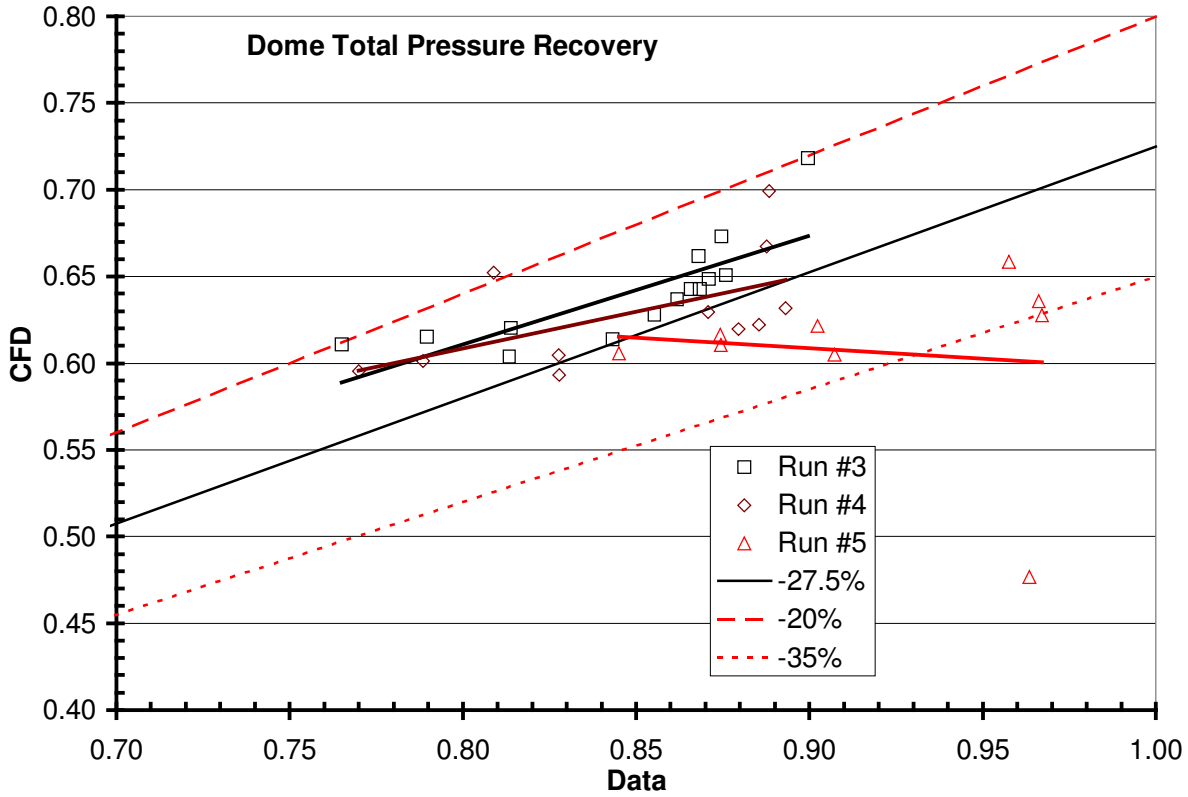


Figure 14: Dome total pressure recovery data compared with 2-D simulation w/ the standard k-ε model with calculated inlet profiles; comparison between Runs 3, 4 and 5.